

RELATIVE WASSERSTEIN ANGLE AND THE PROBLEM OF W_2 -NEAREST GAUSSIAN DISTRIBUTION

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ABSTRACT

Understanding the distributional structure of high-dimensional datasets has become an important topic, yet direct visual characterization is difficult. In this work, we develop a geometric framework for characterizing the distributional structure of empirical datasets by quantifying their deviation from the Gaussian family under the geometry induced by optimal transport theory. Building on the cone structure of the relative translation invariant quadratic Wasserstein (RW_2) space, we define two geometric quantities—the *relative Wasserstein angle* and the *orthogonal projection distance*—and show that they are well-defined because of the flat geometry of the filling cone between distributional rays. This formulation recasts the problem of measuring deviation from the Gaussian family as an orthogonal projection problem onto the Gaussian cone and reveals that the commonly used moment-matching Gaussian is, in general, not the W_2 -nearest Gaussian to a non-Gaussian distribution. In one dimension, we derive closed-form expressions for the proposed quantities and extend closed-form expressions to several other location-scale families, including uniform, Laplace, and logistic distributions. In higher dimensions, we develop a numerical approximation method for the proposed quantities based on empirical optimal transport and covariance-shape optimization. Our experimental results show the empirical convergence and stability of the proposed methods and reveal that the RW_2 angle provides a robust and consistent measure of distributional non-Gaussianity. Moreover, these results provide empirical support for its potential use as an indicator of distributional heterogeneity.

1 INTRODUCTION

With the rapid development of artificial intelligence and data science, datasets have become central to both academic research and industrial applications (Geburu et al., 2021; Paullada et al., 2021). Understanding the structure of a data distribution, including its shape and density, can reveal important aspects of dataset quality (Gong et al., 2023), such as heterogeneity and coverage, which can influence model performance and generalization. Accurately characterizing the distributional structure of datasets has therefore become an important topic in modern data analysis.

Despite its importance, understanding the distributional structure of a dataset through direct visualization is often challenging, particularly when the data lie in a high-dimensional space. In geometry, objects are commonly characterized through comparison with familiar reference shapes, such as spheres or rectangles. Motivated by this perspective, we seek to characterize the distributional structure of high-dimensional datasets by comparing their empirical distributions with standard distribution families. In particular, we focus on the Gaussian family as a typical single-source reference family for assessing distributional structure.

Gaussian distributions are among the most widely used probability distributions in statistics and machine learning. Many models either assume that the underlying data follow a Gaussian distribution or rely on Gaussian noise, priors, or latent variables in machine learning (Bishop, 2006; Arjovsky et al., 2017; Murphy, 2013; Tipping & Bishop, 1999; Kingma & Welling, 2014; Ho et al., 2020; Lipman et al., 2023). Nevertheless, given an empirical distribution, the theoretical study of how

close an empirical distribution is to the Gaussian distribution family remains limited. Classical statistical quantities, such as skewness, kurtosis, and other higher-order moments, can capture specific differences from Gaussianity (Mardia, 1970; Jarque & Bera, 1987; Hall, 1992); however, these quantities can only provide specific assessments and lack a unified geometric framework for quantifying the discrepancy between an empirical distribution and the Gaussian family.

In contrast to this classical statistical perspective, optimal transport (OT) theory provides a natural and unified geometric framework for assessing the distributional difference between two probability distributions (Villani, 2003; 2009; Peyré & Cuturi, 2019). The Wasserstein distance, arising from OT theory, treats probability distributions as points in a metric space and quantifies their discrepancy through the minimum cost of transporting mass from one distribution to another. Because of its intuitive geometric interpretation, optimal transport has become an ideal tool for analyzing distributional differences. This perspective naturally leads to the following question:

From the perspective of optimal transport, given an empirical distribution μ , how close is it to the Gaussian family? Equivalently, to what extent can μ be regarded as Gaussian?

In this paper, we show that this question can be reformulated as a problem of finding the minimal orthogonal projection distance from the distributional point μ to the convex cone of Gaussian distributions under the quadratic Wasserstein geometry. Building on this geometric perspective, we introduce two novel geometric notions, the relative quadratic Wasserstein angle (the RW_2 angle) and the orthogonal projection distance, to quantify the degree of non-Gaussianity for an empirical distribution μ . We further derive closed-form formulae for both quantities in the one-dimensional case and develop an approximation algorithm for higher-dimensional settings. Finally, we conduct two experiments to validate the proposed theories and methods. Our experimental results demonstrate the convergence and stability of the proposed methods and show that the RW_2 angle can provide a robust and consistent measure of distributional non-Gaussianity. Moreover, the results suggest that the RW_2 angle can capture meaningful aspects of distributional heterogeneity.

Organization. The remainder of this paper is organized as follows. Section 2 reviews classical results from optimal transport theory, including the RW_2 metric space and the convex cone of Gaussian distributions. Section 3 introduces the RW_2 angle, the orthogonal projection distance, the filling cone, and derives closed-form expressions and computational algorithms for these quantities. Section 4 presents experimental results that validate our theoretical analyses and methods.

Notations. Let $\mathcal{P}_p(\mathbb{R}^d)$ denote the set of probability distributions on $\mathbb{R}^d$ with finite p th-order moments. For a distribution μ , let $\bar{\mu}$ and Σ_μ denote its mean and covariance matrix, respectively. The moment-matching Gaussian of distribution μ is $\mathcal{N}(\bar{\mu}, \Sigma_\mu)$. We write $[\mu]$ for the translation-equivalence class of μ and $RW_p([\mu], [\nu])$ for the relative translation invariant p -Wasserstein distance between $[\mu]$ and $[\nu]$. The quotient space equipped with the RW_2 metric is denoted by $\mathcal{P}_2(\mathbb{R}^d)/\sim_T$, and $\|[\mu]\|_{RW_2}$ denotes the distance from $[\mu]$ to the equivalence class of the Dirac distribution $[\delta_0]$. The ray induced by positive dilation of $[\mu]$ is denoted by $[[\mu]] := \{(s \text{Id})_\# [\mu] : s > 0\}$. The relative Wasserstein angle between the rays $[[\mu]]$ and $[[\nu]]$ is denoted by $\angle([[\mu]], [[\nu]])$, and $p([\mu], [[\nu]])$ denotes the orthogonal projection distance from $[\mu]$ to $[[\nu]]$. Finally, $\mathcal{N}(0, \Sigma)$ denotes a zero-mean Gaussian distribution with covariance matrix Σ , and $C(\mathcal{N})$ is used to denote the Gaussian cone.

Related Work. Estimating deviation from Gaussianity has a long history in statistics. Classical approaches rely on higher-order moments, such as skewness and kurtosis, to detect asymmetry and tail behavior beyond the Gaussian model (Mardia, 1970; Jarque & Bera, 1987; D’Agostino & Stephens, 1990). Beyond moment-based criteria, information-theoretic measures characterize non-Gaussianity through entropy and divergence. Representative approaches include negentropy (Hyvärinen & Oja, 2000), which exploits the maximum-entropy property of Gaussian distributions, as well as divergence-based measures such as the Kullback–Leibler divergence (Kullback & Leibler, 1951), Rényi divergences, and related quantities in information geometry (Amari, 2016). While powerful and widely used, these approaches are typically tied to specific functionals of the distribution, but do not provide a unified geometric framework for different types of distributions.

Unlike the statistical approaches discussed above, optimal transport and Wasserstein distances measure discrepancies between distributions from a geometric viewpoint (Villani, 2003; 2009; Santambrogio, 2015). Several related works have studied Wasserstein distances between general probability

distributions and Gaussian distributions, with particular focus on covariance structures (Bobkov & Ledoux, 2019; Gelbrich, 1990). A closely related notion is the Bures–Wasserstein angle introduced in van Oostrum (2022), which can be viewed as a special case of the RW_2 angle when restricted to Gaussian distributions. Despite these developments, the moment-matching Gaussian remains a widely used and natural Gaussian approximation. In this work, we show that the moment-matching Gaussian is not, in general, the W_2 -nearest Gaussian to a non-Gaussian distribution. To the best of our knowledge, our work is the first to characterize and quantify the non-Gaussianity of data distributions through geometric measures based on angles and projections in optimal transport.

2 PRELIMINARIES

2.1 OPTIMAL TRANSPORT THEORY

Optimal transport theory provides a geometric framework for comparing probability distributions by minimizing the transportation cost from one measure to another. Let $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$ be two probability measures with finite p -th moments. Given the cost function $c(x, y) = \|x - y\|^p$, the p -norm optimal transport problem is defined as follows.

Definition 2.1 (p -norm Optimal Transport Problem (Villani, 2003)).

$$\text{OT}(\mu, \nu, p) := \min_{\gamma \in \Gamma(\mu, \nu)} \int \|x - y\|^p d\gamma(x, y),$$

where $\Gamma(\mu, \nu)$ denotes the set of all transport plans with marginals μ and ν , namely,

$$\Gamma(\mu, \nu) = \left\{ \gamma \in \mathcal{P}_p(\mathbb{R}^{2d}) \mid \int \gamma(x, y) dy = \mu(x), \int \gamma(x, y) dx = \nu(y) \right\}.$$

The corresponding p -Wasserstein distance is defined by

$$W_p(\mu, \nu) := \text{OT}(\mu, \nu, p)^{1/p}, \quad p \in [1, \infty),$$

which defines a metric on $\mathcal{P}_p(\mathbb{R}^d)$ (Villani, 2009).

Building on this framework, Wang et al. (2026) introduced a translation-invariant extension by identifying probability measures that differ only by a translation. Let $\sim_T$ denote the translation equivalence relation and let $[\mu]$ be the corresponding equivalence class of μ . The p -th order relative translation invariant Wasserstein distance is defined by

$$RW_p([\mu], [\nu]) := \min_{t \in \mathbb{R}^d} W_p(\mu + t, \nu),$$

which is a well-defined metric on the quotient space $\mathcal{P}_p(\mathbb{R}^d)/\sim_T$.

For the quadratic case $p = 2$, the Wasserstein distance admits the decomposition

$$W_2^2(\mu, \nu) = \|\bar{\mu} - \bar{\nu}\|_2^2 + RW_2^2([\mu], [\nu]), \quad (1)$$

where $\bar{\mu}$ and $\bar{\nu}$ denote the means of μ and ν , respectively. Consequently, RW_2 distance measures the intrinsic discrepancy between distributions after removing the translational difference. More importantly, it induces a new metric space of probability distributions, $(\mathcal{P}_2(\mathbb{R}^d)/\sim_T, RW_2)$, which we refer to as the RW_2 metric space and study throughout the remainder of this paper.

2.2 THE GAUSSIAN CONVEX CONE IN THE RW_2 SPACE

As shown by Takatsu (2011), the metric completion of zero-mean Gaussian distributions under the W_2 distance forms a convex cone:

$$C(\mathcal{N}) := \{\mathcal{N}(0, \Sigma) \mid \Sigma \in \text{Sym}_*^d\},$$

where Sym_*^d denotes the set of symmetric nonnegative semi-definite $d \times d$ covariance matrices.

The apex of this cone is the Dirac measure δ_0 , corresponding to the degenerate Gaussian distribution with zero covariance. For any $\Sigma \in \text{Sym}_*^d$ and $s > 0$, scaling the covariance matrix gives

$$\mathcal{N}(0, s^2 \Sigma) = (s \text{Id})_{\#} \mathcal{N}(0, \Sigma),$$

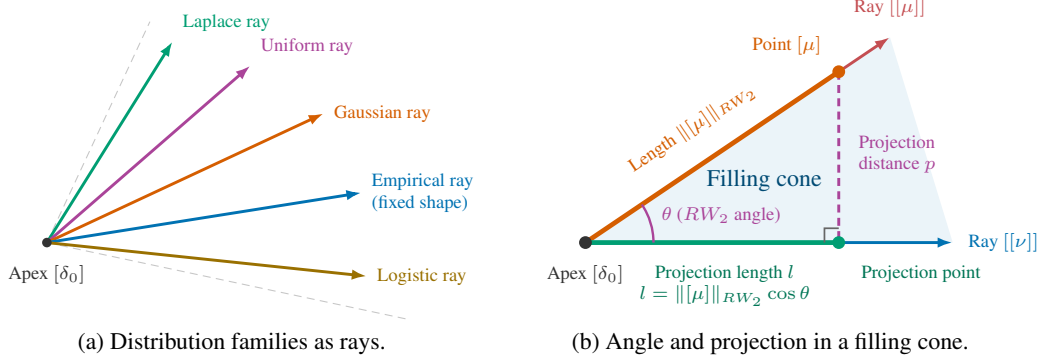


Figure 1: Schematic illustration of the cone geometry of the RW_2 space in the one-dimensional case. (a) Positive rescalings of a fixed distribution form a ray emanating from the apex $[\delta_0]$. The labeled rays illustrate representative distribution families. (b) The shaded region represents the *filling cone* between the rays $[[\mu]]$ and $[[\nu]]$, and their relative Wasserstein angle is θ . The point $[\mu]$ lies at distance $[[\mu]]_{RW_2}$ from the apex. Its orthogonal projection onto $[[\nu]]$ has radial distance $l = [[\mu]]_{RW_2} \cos \theta$, while the dashed segment indicates the projection distance p .

where $(s \text{Id})_\#$ is the push-forward induced by the dilation map $x \mapsto sx$. Therefore, for each nonzero covariance matrix Σ , the family $\{\mathcal{N}(0, s^2 \Sigma) : s > 0\}$ forms a geodesic ray emanating from δ_0 in the RW_2 space. The matrix Σ determines the shape of the Gaussian distribution, while s controls its scale along the ray. As a result, the family of Gaussian distributions naturally admits a convex cone in the RW_2 space.

3 METHODOLOGY

In this section, we introduce our main theory. We first introduce the cone geometry of the RW_2 metric space and define the relative Wasserstein angle, or RW_2 angle. By showing that the filling cone between two distributional rays has zero sectional curvature, we establish that orthogonal projections and an inner-product-like quantity are also well-defined. We then show that both the RW_2 angle and the projection distance can be used to quantify the degree of non-Gaussianity for empirical distributions. Finally, we present algorithms for computing the RW_2 angle and the projection distance in both one-dimensional and high-dimensional settings.

3.1 RELATIVE WASSERSTEIN ANGLES AND ORTHOGONAL PROJECTIONS

Cone Geometry. As mentioned in Section 2.1, RW_2 defines a metric on $\mathcal{P}_2(\mathbb{R}^d)/\sim_T$, thereby yielding a metric space, denoted by $(\mathcal{P}_2(\mathbb{R}^d)/\sim_T, RW_2)$. Compared with the space $(\mathcal{P}_2(\mathbb{R}^d), W_2)$, a distinctive feature of this quotient space is its cone geometry. Specifically, all Dirac distributions in $(\mathcal{P}_2(\mathbb{R}^d), W_2)$ collapse into a single equivalence class $[\delta_0]$, which serves as the *apex* (the origin) of the conic space $(\mathcal{P}_2(\mathbb{R}^d)/\sim_T, RW_2)$. Other distributions form *rays* emanating from this apex, each corresponding to a distributional “shape” up to translation and positive scaling.

To formalize this observation, we introduce the *dilation (positive scaling) equivalence relation* $\sim_S$ on the space $\mathcal{P}_2(\mathbb{R}^d)/\sim_T$, defined by

$$[\mu] \sim_S [\nu] \iff [\nu] = (s \text{Id})_\# [\mu] \text{ for some } s > 0.$$

This relation partitions $\mathcal{P}_2(\mathbb{R}^d)/\sim_T \setminus \{[\delta_0]\}$ into *disjoint rays*. For each non-apex distributional point $[\mu]$, the associated ray $[[\mu]]$ is defined as the collection of all positive dilations of $[\mu]$,

$$[[\mu]] := \{(s \text{Id})_\# [\mu] : s > 0\}.$$

Here, the double-bracket notation $[[\cdot]]$ is used to reflect two quotients by translation and dilation.

Geometrically, each ray $[[\mu]]$ is a geodesic in the RW_2 metric space, as the shortest path between any two points on the ray is obtained by continuously varying the dilation s . Consequently, the space $(\mathcal{P}_2(\mathbb{R}^d)/\sim_T, RW_2)$ admits a cone structure, consisting of an apex and rays generated by dilations.

Relative Wasserstein Angle. Inspired by the notion of angles in metric spaces (Ambrosio et al., 2005), we introduce an angle concept on this quotient space.

Definition 3.1 (Relative Wasserstein Angle). *For any two rays $[[\mu]], [[\nu]] \in \mathcal{P}_2(\mathbb{R}^d)/\sim_T$, the relative Wasserstein angle between them, denoted by the RW_2 angle, is defined as*

$$\angle([[\mu]], [[\nu]]) := \arccos \frac{a^2 + b^2 - c^2}{2ab},$$

where $a = RW_2([\mu], [\delta_0])$, $b = RW_2([\nu], [\delta_0])$, and $c = RW_2([\mu], [\nu])$.

This definition follows the law of cosines and yields a well-defined angle notion between rays in $(\mathcal{P}_2(\mathbb{R}^d)/\sim_T, RW_2)$. The resulting angle quantifies the *directional* difference between two rays and is invariant under positive rescaling along each ray.

Orthogonal Projection. The classical Wasserstein space is known to be an Alexandrov space of *nonnegative* curvature, and thus Euclidean geometric properties generally do not hold globally (Ambrosio et al., 2005; Takatsu, 2011). Nevertheless, we show that the relative Wasserstein angle behaves like a Euclidean angle, since it can be embedded in a two-dimensional Euclidean plane.

Given two rays in the RW_2 space, we define the *filling cone* between them as the union of all geodesics connecting points on the two rays, as illustrated in Figure 1b. The following result characterizes the geometry of this filling cone.

Theorem 3.2. *If at least one of the two distinct non-apex rays is generated by an absolutely continuous probability distribution¹, then the corresponding filling cone has zero sectional curvature. Moreover, it is isometric to a cone region in the two-dimensional Euclidean plane.*

The proof is provided in Appendix B. This isometric property of the filling cone allows Euclidean notions, such as angles and projections, to be well-defined in the probability distribution space.

For projections, we use the closed ray $\overline{[[\nu]]} := [[\nu]] \cup \{[\delta_0]\} = \{(s \text{Id})_{\#}[\nu] : s \geq 0\}$, which includes the apex to ensure that the projection is attained even when the angle between the rays equals $\pi/2$.

Definition 3.3 (Projection Distance). *Given a non-apex point $[\mu]$ and a ray $[[\nu]]$, the projection distance from point $[\mu]$ to ray $[[\nu]]$ is defined as*

$$p([\mu], [[\nu]]) := \min_{[\eta] \in [[\nu]]} RW_2([\mu], [\eta]) = \|\mu\|_{RW_2} \sin \angle([[\mu]], [[\nu]]).$$

For the apex, we set $p([\delta_0], [[\nu]]) = 0$.

Here, $\|\mu\|_{RW_2} := RW_2([\mu], [\delta_0])$ denotes the *length* of $[\mu]$, the distance to the apex $[\delta_0]$.

Similarly, we may also define an inner-product-like quantity between distribution equivalence classes by

$$\langle [\mu], [\nu] \rangle := \|\mu\|_{RW_2} \|\nu\|_{RW_2} \cos \angle([[\mu]], [[\nu]]).$$

This notion can be interpreted as an average inner product induced by the optimal couplings, though this quantity does not make the quotient space an inner product space, since its underlying geometry is conic rather than linear. Appendix G gives an example in which three rays have unique flat pairwise filling cones, but the union of these cones, equipped with the RW_2 metric, does not admit a Euclidean isometric embedding. Thus, pairwise flatness does not extend trivially to three or more rays.

3.2 W_2 -NEAREST GAUSSIAN DISTRIBUTION

Having introduced these geometric notions, we now turn to the central question: given an empirical distribution μ , to what extent can it be regarded as Gaussian?

From the perspective of optimal transport, this question can be viewed as finding the *minimal* W_2 distance between the distribution μ and the Gaussian family. By the mean–variance decomposition in Equation 1, any Gaussian minimizer of the W_2 distance, which we refer to as a W_2 -nearest Gaussian, must have the same mean as μ . Consequently, the problem can be further reduced to optimizing over the covariance matrices of Gaussian distributions.

¹The absolute continuity condition guarantees the uniqueness of the optimal transport plan and, consequently, the uniqueness of the corresponding filling cone.

Projection onto the Gaussian Cone. As discussed in Section 2.2, the Gaussian family admits a convex cone $C(\mathcal{N})$ in the RW_2 metric space, including both the apex and Gaussian rays determined by covariance shapes. As the minimal W_2 distance must be a projection distance from the point $[\mu]$ to a certain ray within the Gaussian cone $C(\mathcal{N})$, the central problem of finding the minimal W_2 distance from distribution μ to a Gaussian distribution can be reformulated as finding the projection distance from the point $[\mu]$ onto the Gaussian cone $C(\mathcal{N})$ in the metric space $(\mathcal{P}_2(\mathbb{R}^d)/\sim_T, RW_2)$. Moreover, because the projection distance onto a ray is determined by the relative Wasserstein angle, this problem can also be formulated as finding the minimum RW_2 angle between the ray $[[\mu]]$ and a Gaussian ray in $C(\mathcal{N})$.

Solutions in One and Higher Dimensions. In the one-dimensional case, the Gaussian family corresponds to a single ray in the RW_2 space. This geometric structure yields closed-form expressions for both the RW_2 angle and the projection distance, as detailed in Section 3.3.

In higher dimensions, the problem generally does not admit a closed-form solution, except for certain specially structured distributions, such as coordinate-separable distributions in a principal component analysis (PCA) basis; see Appendix F. For the general case, we develop a numerical approximation method to solve the problem, as described in Section 3.4.

The Moment-Matching Gaussian. A commonly used Gaussian approximation of μ in statistics is the moment-matching Gaussian distribution $\mu_{\text{MM}} = \mathcal{N}(\bar{\mu}, \Sigma_\mu)$, where $\bar{\mu}$ and Σ_μ denote the mean and covariance of μ , respectively. It is natural to conjecture that this distribution is the W_2 -nearest Gaussian to μ . However, using the proposed notions of angle and projection, we show that the moment-matching Gaussian μ_{MM} is *not* the W_2 -nearest Gaussian to a non-Gaussian μ .

Here, we provide a proof sketch, while the detailed derivation is given in Appendix E. Since μ_{MM} shares the same covariance as μ , the two distributions have the same RW_2 length:

$$\|[\mu_{\text{MM}}]\|_{RW_2} = \|[\mu]\|_{RW_2} = \sqrt{\text{Tr}(\Sigma_\mu)}.$$

However, when μ is non-Gaussian, the RW_2 angle between the ray $[[\mu]]$ and the Gaussian ray $[[\mu_{\text{MM}}]]$ is strictly positive. Consequently, the orthogonal projection of $[\mu]$ onto this Gaussian ray has a smaller length than $[\mu_{\text{MM}}]$. Therefore, μ_{MM} cannot be the W_2 -nearest Gaussian to μ .

Moreover, we also find that, for certain distributions μ , the W_2 -nearest Gaussian may share neither the same eigenvalues nor the same eigenvectors of the covariance matrix with the moment-matching Gaussian; an example is provided in Appendix H. These examples suggest that analytic solutions are unlikely to exist in general for higher-dimensional cases. Therefore, we adopt a numerical approximation method, which is discussed in Section 3.4.

3.3 CLOSED-FORM SOLUTIONS IN ONE DIMENSION

In the one-dimensional case, all nondegenerate zero-mean Gaussian distributions lie on a single ray in the RW_2 metric space, namely $[[\mathcal{N}(0, \sigma^2)]]$, with $\sigma > 0$. Exploiting the monotone optimal transport map in one dimension, we derive closed-form expressions for both the RW_2 angle and the corresponding projection distance, as stated in Proposition 3.4. The detailed derivation is provided in Appendix C, and Algorithm 1 in Appendix A summarizes the computational procedure.

Proposition 3.4 (Closed-form solutions in one dimension). *Let μ be a zero-mean empirical distribution on $\mathbb{R}$ with length $\sigma_\mu > 0$, given by $\mu := \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$, $\bar{\mu} = \frac{1}{n} \sum_{i=1}^n x_i = 0$, $\sigma_\mu = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}$, where $x_1 \leq \dots \leq x_n$ are empirical samples. Let ϕ and Φ denote the probability density function and cumulative distribution function of the standard normal distribution $\mathcal{N}(0, 1)$, respectively. Define*

$$z_0 \leftarrow -\infty, \quad z_i \leftarrow \Phi^{-1}\left(\frac{i}{n}\right), \quad i = 1, \dots, n-1, \quad z_n \leftarrow +\infty.$$

Define the projection length $l := \sum_{i=1}^n x_i (\phi(z_{i-1}) - \phi(z_i))$. Then for the ray $[[\mu]]$ and the Gaussian ray $[[\mathcal{N}(0, \sigma^2)]]$, their relative Wasserstein angle θ and the corresponding projection distance p from $[\mu]$ to this Gaussian ray, are given by

$$\theta = \arccos\left(\frac{l}{\sigma_\mu}\right), \quad p = \sigma_\mu \sin \theta. \quad (2)$$

Remark. The above formulae connect directly to classical work on Wasserstein-based tests of normality (Brown & Hettmansperger, 1996; del Barrio et al., 1999). In particular, del Barrio et al. (1999) derived the normalized squared distance to the Gaussian family. In our notation, their statistic is $R_n = \frac{p^2}{\sigma_\mu^2} = \sin^2 \theta$. Thus, the one-dimensional RW_2 angle is a monotone transformation of this existing statistic. Our contribution here is to interpret these quantities as an angle and an orthogonal projection distance in the cone geometry of the RW_2 space, within a framework that also applies in higher dimensions.

Complexity Analysis. The computation requires $\mathcal{O}(n \log n)$ time, dominated by sorting the empirical samples, and $\mathcal{O}(n)$ memory.

The same approach can be extended to other distribution families parametrized by mean and scale, commonly referred to as *location–scale* families, such as the uniform, Laplace, and logistic distributions. Their corresponding formulae are provided in Appendix D. For the distribution families that are not location–scale cases, such as geometric distributions, the problem becomes one of projecting onto more general curves in the RW_2 space. We leave this direction for future work.

3.4 SOLUTIONS IN HIGHER DIMENSIONS

Unlike the one-dimensional case, in which the Gaussian family forms a single ray in the RW_2 metric space, the Gaussian family in higher dimensions consists of multiple rays determined by their covariance matrices. Consequently, finding the W_2 -nearest Gaussian to a given distribution μ requires not only computing the projection distance from $[\mu]$ onto a fixed Gaussian ray, but also optimizing over the space of Gaussian covariance matrices.

Concretely, for a given distribution μ , the central problem is given by

$$\min_{\Sigma \in \text{Sym}_*^d} \min_{s > 0} RW_2^2([\mu], [\mathcal{N}(0, s^2 \Sigma)]), \quad (3)$$

where Sym_*^d denotes the set of symmetric nonnegative semi-definite covariance matrices, and s controls the scale along the Gaussian ray associated with Σ .

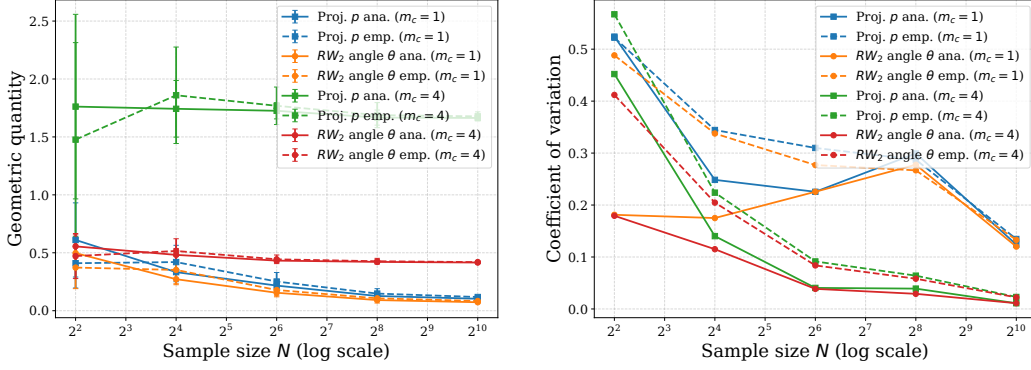
We can parameterize the covariance matrix through its eigendecomposition, $\Sigma = U \Lambda U^\top$, where $U \in \mathcal{O}(d)$ is an orthogonal matrix and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ contains the nonnegative eigenvalues $\lambda_i \geq 0$. Since the Gaussian ray depends only on the covariance shape, we can reduce the scale parameter s by imposing the normalization $\text{tr}(\Sigma) = \sum_{i=1}^d \lambda_i = 1$.

Under these parameter reductions, Problem 3 simplifies to unit-trace optimization

$$\min_{U \in \mathcal{O}(d), \Lambda \succeq 0, \text{Tr}(\Lambda)=1} RW_2^2([\mu], [\mathcal{N}(0, U \Lambda U^\top)]).$$

For numerical computation, we approximate the continuous Gaussian distribution $\mathcal{N}(0, U \Lambda U^\top)$ using samples $\{y_i\}_{i=1}^m \sim \mathcal{N}(0, U \Lambda U^\top)$. We initialize the samples by drawing from the normalized Gaussian $\mathcal{N}(0, \frac{1}{d} I_d)$. At each iteration, the Gaussian samples are updated alternately between an optimal-transport gradient step and a retraction onto the normalized Gaussian covariance constraint. Here, the gradient update is computed from the optimal transport coupling between the current Gaussian samples and empirical samples from μ . After each gradient update, we estimate the covariance matrix of the centered samples, normalize its eigenvalues to have unit sum, and draw new Gaussian samples from the resulting covariance matrix. This procedure is repeated until the stopping criterion is met or the iteration limit is reached. Once an optimal Gaussian covariance shape Σ^* with $\text{tr}(\Sigma^*) = 1$ and its corresponding RW_2 angle θ^* are obtained, a W_2 -nearest Gaussian can be recovered by $\nu^* = \mathcal{N}(\bar{\mu}, r^2 \cos^2 \theta^* \Sigma^*)$, $r = \|\mu\|_{RW_2} = \sqrt{\text{tr}(\Sigma_\mu)}$. The complete algorithm and its complexity analysis are provided in Appendix A.2. Figure 5 further shows the convergence behavior and stability of the proposed algorithm.

Remarks. In low-dimensional settings, semi-discrete optimal transport methods based on power diagrams, also known as Laguerre diagrams, can provide more accurate estimates of the RW_2 distance by avoiding the sampling approximation of the continuous Gaussian distribution (Peyré & Cuturi, 2019). However, these methods become computationally prohibitive in high-dimensional



(a) RW_2 angle and projection distance as measurements of non-Gaussianity. (b) Comparison of coefficients of variation (CV).

Figure 2: Comparison of the RW_2 angle θ and projection distance p for quantifying the non-Gaussianity of a Gaussian mixture distribution. (a) Both quantities decrease as the mixture separation decreases, reflecting reduced non-Gaussianity. As the sample size N increases, the sampling-based estimates converge to the corresponding closed-form values. (b) Coefficients of variation, $CV = \text{std}/|\text{mean}|$, computed over 10 independent repetitions. The RW_2 angle consistently exhibits a lower CV than the projection distance, indicating it is more robust than the projection distance.

settings, such as $d = 512$, due to the cost of constructing and integrating over Laguerre cells. In addition, very small covariance eigenvalues can lead to ill-conditioned covariance matrices, making the numerical optimization sensitive to gradient updates. In practice, it is also useful to incorporate PCA-based dimensionality reduction to reduce computational cost and improve numerical stability.

4 EXPERIMENTS

In this section, we conduct two experiments to validate the proposed theoretical analysis and demonstrate the practical utility of our framework and algorithms. The first experiment focuses on the one-dimensional case, where the theoretical analyses can be examined directly. The second experiment investigates a practical application for characterizing the distributional structure of high-dimensional datasets, including aspects of heterogeneity. All experiments are conducted on a Linux workstation equipped with an Intel Core i7 CPU, 16 GB of RAM, and an NVIDIA RTX 3090 GPU.

4.1 NON-GAUSSIANITY EVALUATION IN ONE DIMENSION

Setup. We demonstrate the ability of the proposed geometric quantities to characterize non-Gaussianity in one dimension. We consider the following equally weighted Gaussian mixture,

$$\mu = \frac{1}{2}\mathcal{N}(-m_c, 1) + \frac{1}{2}\mathcal{N}(m_c, 1),$$

where $m_c \in \{1, 4\}$ controls the separation between the two component means. For each sample size $N \in \{2^2, 2^4, 2^6, 2^8, 2^{10}\}$, we draw N samples from μ to construct an empirical distribution $\hat{\mu}_N$.

We consider two methods for computing the RW_2 angle θ and projection distance p . The first method uses the closed-form expressions in Algorithm 1, treating the Gaussian distribution continuously. The second one approximates the Gaussian distribution using N samples. We repeat each experiment independently 10 times and report the mean and standard deviation of both quantities. To compare the robustness of the two quantities across repetitions, we also compute the coefficient of variation, $CV = \frac{\text{std}}{|\text{mean}|}$, where smaller values indicate better robustness.

Results. Figure 2a shows that both the RW_2 angle and the projection distance decrease as m_c decreases, indicating that both quantities can capture the reduction in non-Gaussianity. As the sample size N increases, the sampling-based estimates converge to the corresponding closed-form values.

These results show that the proposed quantities can be used to evaluate the non-Gaussianity. Figure 2b reports the coefficient of variation (CV) of the two quantities. We find that the RW_2 angle consistently exhibits a lower CV than the projection distance. This suggests that the angle is more robust than the projection distance, potentially since the former measures differences between rays rather than individual points. In addition, we also include a visual comparison between the moment-matching Gaussian and the W_2 -nearest Gaussian in Appendix I.1.

4.2 NON-GAUSSIANITY EVALUATION ON HIGH-DIMENSIONAL DATASETS

Beyond the one-dimensional setting, we evaluate the proposed approximation algorithm (Algorithm 2) and investigate a practical application for characterizing the distributional structure of high-dimensional datasets. Due to space limitations, we summarize the essential experimental setup and key results here and defer detailed configurations, results, and analyses to Appendix I.2.

Setup. We evaluate both raw and latent data representations. Specifically, we use MNIST (Lecun et al., 1998) and EMNIST (Cohen et al., 2017) for evaluation in pixel space, and CIFAR-10 (Krizhevsky & Hinton, 2009) and CINIC-10 (Darlow et al., 2018) for evaluation in feature space, with features extracted by a pretrained Inception-v3 encoder (Szegedy et al., 2015).

For each dataset, we consider the full empirical distribution, individual classes, and selected mixtures of classes. We use Algorithm 2 to estimate the RW_2 angle between each distribution and its W_2 -nearest Gaussian approximation, with the moment-matching Gaussian approximation as a baseline. We report the mean and standard deviation of the estimated angles and running times. Each experiment is repeated five times with different random seeds. We also track the RW_2 angle over optimization iterations to assess the empirical convergence and stability of the proposed algorithm.

Results. Figure 5 shows the evolution of the estimated RW_2 angle between each MNIST digit-class distribution and its Gaussian approximation during the first experimental repeat. Each curve corresponds to one digit class (0–9). The angles decrease rapidly during the initial iterations and subsequently stabilize around class-dependent values, providing empirical evidence of the convergence and stability of Algorithm 2.

Across MNIST, EMNIST, CIFAR-10, and CINIC-10, the estimated W_2 -nearest Gaussian consistently yields smaller RW_2 angles than the moment-matching baseline, at a higher computational cost. The full empirical distributions generally exhibit larger angles than most individual classes, and the angles increase as more classes are combined in the selected mixture sequences. Although paired datasets exhibit similar angles for their full distributions, differences emerge at the class level, such as digit 4 in MNIST versus EMNIST and the horse class in CIFAR-10 versus CINIC-10. Together, these results demonstrate that the RW_2 angle captures differences in non-Gaussianity across datasets, classes, and mixtures in both pixel-based and latent representations, providing empirical support for its potential use as an indicator of distributional heterogeneity.

5 CONCLUSION

In this paper, we introduced a geometric framework for quantifying deviations from the Gaussian family under Wasserstein geometry. By exploiting the cone structure of the RW_2 space and the flatness of the filling cone, we established that the relative Wasserstein angle and orthogonal projection distance are well-defined and provide interpretable measures of non-Gaussianity. We derived closed-form expressions in one dimension and extended the analysis to several other location-scale families. For high-dimensional distributions, we developed approximation algorithms based on empirical optimal transport, with PCA used to reduce computational cost. We also demonstrated that the moment-matching Gaussian is generally *not* the W_2 -nearest Gaussian to a non-Gaussian distribution.

Experiments on synthetic distributions and high-dimensional datasets validate the proposed framework. Both the RW_2 angle and projection distance capture deviations from Gaussianity, with the angle exhibiting greater robustness to sampling variability. The RW_2 angle also reveals differences in non-Gaussianity across datasets, classes, and mixtures in both pixel-based and latent representations, providing empirical support for its potential use as an indicator of distributional heterogeneity.

More broadly, these quantities offer a geometric perspective on Wasserstein discrepancy beyond distance alone. Promising directions for future work include dataset characterization, distribution shift analysis, and applications to flow matching and diffusion models, where the geometry of probability distributions plays a central role.

ETHICS STATEMENT

This work develops mathematical and computational methods for characterizing probability distributions and evaluates them using established benchmark datasets, including MNIST, EMNIST, CIFAR-10, and CINIC-10. The study does not involve new human-subject experiments, interventions, or the collection of private or sensitive personal information. We use existing datasets and pretrained models in accordance with their intended research use and appropriately cite their original sources.

A potential concern is the interpretation of the proposed RW_2 angle as a general measure of dataset quality. The RW_2 angle specifically quantifies deviation from a reference distribution family, the Gaussian family in this work, under a chosen representation and experimental protocol. It should therefore not be interpreted by itself as a measure of fairness, social representativeness, completeness, or overall dataset quality. In particular, measurements in latent feature space depend on the selected feature extractor, dimensionality-reduction procedure, sampling strategy, and other preprocessing choices. We consequently restrict our conclusions to the geometric properties measured under the stated experimental settings and do not advocate using the proposed metric alone for consequential dataset selection or auditing decisions. The authors have read and agree to adhere to the ICLR Code of Ethics.

REPRODUCIBILITY STATEMENT

We have taken several steps to support reproducibility of both the theoretical and experimental results. The definitions, assumptions, and principal theoretical results are presented in Sections 2 and 3, with detailed derivations and proofs provided in the corresponding appendices. Experimental protocols are described in Section 4, including the datasets, representations, preprocessing procedures, PCA dimensionality reduction, sampling strategy, optimization procedure, evaluation settings, and reported repetitions. An anonymized implementation and the corresponding experimental configuration files are provided as supplementary material to facilitate reproduction of the numerical results.

AI USE STATEMENT

Generative AI tools, including ChatGPT, Claude and DeepSeek, were used during the preparation of this work to assist with refining mathematical statements and their exposition, providing feedback on experimental design, assisting with the implementation and debugging of experimental code, and presenting the experimental results. Generative AI tools were also used to improve the clarity and readability of the manuscript, assist with manuscript organization, summarize and identify relevant literature, and help with reference formatting and related writing tasks. We did not use generative AI to generate the experimental datasets used in this work; translation and qualitative or thematic data analysis are not applicable to this study.

All AI-assisted content was reviewed by the authors. In particular, the mathematical definitions, assumptions, derivations, theoretical claims, and proofs were independently checked by the authors; AI-assisted code was reviewed, executed, and tested; numerical results were obtained from the authors' experimental implementations; and references and literature-related information were checked against the original sources where appropriate. Generative AI was not treated as an authoritative source for mathematical or scientific claims.

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A ALGORITHMS

A.1 RW_2 ANGLE AND PROJECTION DISTANCE IN ONE DIMENSION

Algorithm 1 RW_2 Angle and Projection Distance in One Dimension

Require: Samples $\{x_i\}_{i=1}^n \subset \mathbb{R}$

Ensure: RW_2 angle θ and projection distance p

- 1: Compute the sample mean: $\bar{x} \leftarrow \frac{1}{n} \sum_{i=1}^n x_i$
- 2: Center the samples: $x_i \leftarrow x_i - \bar{x}, \quad i = 1, \dots, n$
- 3: Sort the centered samples so that $x_1 \leq \dots \leq x_n$
- 4: Compute the empirical standard deviation: $\sigma_\mu \leftarrow \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}$
- 5: Set the Gaussian quantile boundaries:

$$z_0 \leftarrow -\infty, \quad z_i \leftarrow \Phi^{-1}\left(\frac{i}{n}\right), \quad i = 1, \dots, n-1, \quad z_n \leftarrow +\infty$$

- 6: Compute the projection length:

$$l \leftarrow \sum_{i=1}^n x_i (\phi(z_{i-1}) - \phi(z_i))$$

- 7: Compute the angle: $\theta \leftarrow \arccos(l/\sigma_\mu)$
 - 8: Compute the projection distance: $p \leftarrow \sigma_\mu \sin \theta$
 - 9: **return** θ, p
-

Here, Φ and ϕ denote the cumulative distribution function and probability density function of the standard normal distribution, respectively, with $\phi(-\infty) = \phi(+\infty) = 0$. If the empirical variance is zero, the distribution corresponds to the apex of the RW_2 space; its projection distance is zero, while its angle is undefined.

A.2 NUMERICAL APPROXIMATION OF THE RW_2 ANGLE IN HIGH DIMENSIONS**Algorithm 2** Numerical Approximation of the RW_2 Angle in High Dimensions

Require: Samples $\{x_i\}_{i=1}^n \subset \mathbb{R}^d$, Gaussian sample size m , step size η , tolerance ε , maximum number of iterations $T_{\max}$

Ensure: Estimated covariance shape Σ^* and RW_2 angle θ^*

- 1: Center the input samples: $x_i \leftarrow x_i - \frac{1}{n} \sum_{k=1}^n x_k$
- 2: Initialize $\Sigma^{(0)} \leftarrow I_d/d$
- 3: Draw $y_1^{(0)}, \dots, y_m^{(0)} \sim \mathcal{N}(0, \Sigma^{(0)})$
- 4: Center the Gaussian samples: $y_j^{(0)} \leftarrow y_j^{(0)} - \frac{1}{m} \sum_{k=1}^m y_k^{(0)}$
- 5: set $\theta^{(0)} \leftarrow +\infty$
- 6: **for** $t = 1, \dots, T_{\max}$ **do**
- 7: Compute the optimal transport coupling with uniform marginals:

$$\gamma^{(t)} \leftarrow \text{OT}(\{x_i\}_{i=1}^n, \{y_j^{(t-1)}\}_{j=1}^m)$$

- 8: Compute the sample gradients:

$$g_j^{(t)} \leftarrow \sum_{i=1}^n \gamma_{ij}^{(t)} (y_j^{(t-1)} - x_i), \quad j = 1, \dots, m$$

- 9: Update the samples: $\tilde{y}_j^{(t)} \leftarrow y_j^{(t-1)} - \eta g_j^{(t)}$
- 10: Center the updated samples: $\tilde{y}_j^{(t)} \leftarrow \tilde{y}_j^{(t)} - \frac{1}{m} \sum_{k=1}^m \tilde{y}_k^{(t)}$
- 11: Estimate the covariance matrix:

$$\hat{\Sigma}^{(t)} \leftarrow \frac{1}{m} \sum_{j=1}^m \tilde{y}_j^{(t)} (\tilde{y}_j^{(t)})^\top$$

- 12: Compute the eigendecomposition:

$$\hat{\Sigma}^{(t)} = U^{(t)} \text{diag}(\hat{\lambda}_1^{(t)}, \dots, \hat{\lambda}_d^{(t)}) (U^{(t)})^\top$$

- 13: Normalize the eigenvalues to enforce unit trace:

$$\lambda_i^{(t)} \leftarrow \frac{\hat{\lambda}_i^{(t)}}{\sum_{k=1}^d \hat{\lambda}_k^{(t)}}, \quad i = 1, \dots, d$$

- 14: Set $\Sigma^{(t)} \leftarrow U^{(t)} \text{diag}(\lambda_1^{(t)}, \dots, \lambda_d^{(t)}) (U^{(t)})^\top$
- 15: Draw $y_1^{(t)}, \dots, y_m^{(t)} \sim \mathcal{N}(0, \Sigma^{(t)})$
- 16: Center the Gaussian samples: $y_j^{(t)} \leftarrow y_j^{(t)} - \frac{1}{m} \sum_{k=1}^m y_k^{(t)}$
- 17: Compute $\theta^{(t)} \leftarrow \theta_{RW_2}(\{x_i\}_{i=1}^n, \{y_j^{(t)}\}_{j=1}^m)$
- 18: **if** $|\theta^{(t)} - \theta^{(t-1)}| < \varepsilon$ **then**
- 19: **break**
- 20: **end if**
- 21: **end for**
- 22: **return** $\Sigma^* \leftarrow \Sigma^{(t)}, \theta^* \leftarrow \theta^{(t)}$

Here, the factor of 2 in the sample gradients has been absorbed into the step size.

Complexity Analysis. At each iteration, the main computational cost arises from solving the discrete optimal transport problem between n empirical samples and m Gaussian samples in $\mathbb{R}^d$. Constructing the pairwise cost matrix requires $O(nmd)$ operations, while the complexity of computing the transport coupling depends on the chosen OT solver. For example, an entropic OT solver with K Sinkhorn iterations requires $O(Knm)$ operations once the cost matrix is available (Cuturi, 2013; Peyré & Cuturi, 2019). Computing the sample gradients costs $O(nmd)$. Covariance estimation and

Gaussian resampling each require $O(md^2)$ operations, while eigendecomposition costs $O(d^3)$. Thus, when using the Sinkhorn solver, the overall per-iteration complexity is

$$O(nm(d + K) + md^2 + d^3).$$

For T outer iterations, the total complexity is

$$O(T(nm(d + K) + md^2 + d^3)).$$

For large sample sizes and moderate dimension, the transport-related computations typically dominate the runtime. We also report runtime in our experiments to illustrate the practical computational cost of the proposed algorithm.

B PROOF OF FLATNESS OF THE FILLING CONE

In this section, we prove that the two-dimensional filling cone generated by two rays emanating from the apex $[\delta_0]$ in the quotient space $(\mathcal{P}_2(\mathbb{R}^d)/\sim_T, RW_2)$ is flat whenever one of the rays is generated by an absolutely continuous distribution. The key observation is that, after a unique optimal alignment ensured by Brenier’s theorem (Brenier, 1991; Villani, 2003), all Wasserstein distances between any pair of points within the filling cone are induced by a fixed quadratic form in a global affine coordinate system.

B.1 SETUP AND NOTATION

Let $[\mu], [\nu] \in \mathcal{P}_2(\mathbb{R}^d)/\sim_T$ and assume that ray $[[\mu]]$ is different from ray $[[\nu]]$. Without loss of generality, we assume that both representatives, μ and ν , are centered and normalized to unit RW_2 norm, i.e.,

$$\bar{\mu} = \bar{\nu} = 0, \quad \|\mu\|_{RW_2} = \|\nu\|_{RW_2} = 1.$$

Throughout the proof, we assume that the source distribution μ is absolutely continuous.

B.2 UNIQUENESS OF THE ALIGNED OPTIMAL MAP

Lemma B.1 (Uniqueness of the Optimal Transport Map). *If μ is absolutely continuous, then the optimal transport plan between μ and ν for the quadratic cost is unique and given by $(\text{Id}, T)_\# \mu$, where $T = \nabla \phi$ is the gradient of a convex function.*

Proof. This follows directly from Brenier’s theorem (Brenier, 1991; Villani, 2009). $\square$

B.3 GEODESICS IN THE FILLING CONE

Using the unique optimal map T , any pair of points, ρ_0 and ρ_1 , lying on the two rays can be parametrized by scalars $s_0, s_1 \geq 0$ as

$$\rho_0 = (s_0 \text{Id})_\# \mu, \quad \rho_1 = (s_1 T)_\# \mu.$$

Lemma B.2 (Form of Geodesics). *The Wasserstein geodesic connecting ρ_0 and ρ_1 is given by*

$$\rho_t = (F_t)_\# \mu, \quad F_t(x) = (1-t)s_0x + ts_1T(x), \quad t \in [0, 1].$$

Proof. This follows from McCann’s displacement interpolation, generated by the unique optimal plan $(\text{Id}, T)_\# \mu$, composed with the endpoint scalings $x \mapsto s_0x$ and $y \mapsto s_1y$. The result also trivially extends to the cases where $s_0 = 0$ or $s_1 = 0$. $\square$

B.4 COORDINATE CHART IN THE FILLING CONE

Since the filling cone is defined as the union of all Wasserstein geodesics connecting pairs of points lying on the two rays generated by μ and ν , any point ζ in the filling cone must lie on at least one such geodesic. Consequently, it would have the form of

$$\zeta = (F_t)_\# \mu, \quad F_t(x) = ax + bT(x),$$

where $(a, b) = ((1-t)s_0, ts_1)$ for some $t \in [0, 1]$ and $s_0, s_1 \geq 0$ determined by the endpoints $\rho_0 = (s_0 \text{Id})_\# \mu$ and $\rho_1 = (s_1 T)_\# \mu$.

Since the interpolation map F_t depends linearly on x and $T(x)$, the pair (a, b) can be viewed as the coefficients of (Id, T) . Therefore, any point ζ in the filling cone can be described by the coordinates (a, b) with respect to the basis (Id, T) .

Moreover, this pair (a, b) is uniquely determined, and hence the coordinate description is intrinsically well-defined. Indeed, although a given point ζ in the filling cone may lie on two different geodesics corresponding to distinct choices of (t, s_0, s_1) , the uniqueness of the optimal transport map T guarantees that ζ admits a unique representation of the form $\zeta = (a \text{Id} + bT)_\# \mu$. Consequently, we may define one affine coordinate chart on the filling cone.

Motivated by this observation, for general $a, b \geq 0$ we define push forward $S_{a,b}$ as

$$S_{a,b}(x) := ax + bT(x), \quad \rho_{a,b} := (S_{a,b})_{\#}\mu.$$

The filling cone can then be characterized as the image of the map

$$\Phi : \mathbb{R}_{\geq 0}^2 \rightarrow \mathcal{P}_2(\mathbb{R}^d)/\sim_T, \quad \Phi(a, b) = \rho_{a,b}.$$

B.5 EXACT W_2 DISTANCES INSIDE THE FILLING CONE

We begin by recalling two lemmas on optimal transport plans, and subsequently derive the distance formula for any pair of points in the filling cone.

Lemma B.3 (Optimality and c -cyclical Monotonicity). *Let $c : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be continuous and let $\pi \in \Pi(\mu, \nu)$ be a transport plan. Then π is optimal for the cost c if and only if the support is c -cyclically monotone, namely for any finite family*

$$(x_1, y_1), \dots, (x_m, y_m) \in \text{supp}(\pi)$$

one has

$$\sum_{i=1}^m c(x_i, y_i) \leq \sum_{i=1}^m c(x_i, y_{i+1}), \quad y_{m+1} = y_1.$$

Lemma B.4 (Flip Invariance for Quadratic Cost). *Let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be an optimal transport map for the quadratic cost $c(x, y) = \|x - y\|^2$. If the graph*

$$\Gamma = \{(x, T(x))\}$$

is c -cyclically monotone, then the flipped set

$$\Gamma^\top = \{(T(x), x)\}$$

is also c -cyclically monotone.

Proof. For quadratic cost

$$c(x, y) = \|x - y\|^2 = |x|^2 + |y|^2 - 2\langle x, y \rangle.$$

Thus c -cyclical monotonicity of Γ is equivalent to

$$\sum_{i=1}^m \langle x_i, T(x_i) \rangle \geq \sum_{i=1}^m \langle x_i, T(x_{i+1}) \rangle.$$

By cyclic relabeling,

$$\sum_{i=1}^m \langle T(x_i), x_i \rangle \geq \sum_{i=1}^m \langle T(x_i), x_{i+1} \rangle,$$

which is exactly c -cyclical monotonicity of $\Gamma^\top$. $\square$

Proposition B.5 (Distance Induced by Affine Coordinates). *For any $(a_1, b_1), (a_2, b_2) \in \mathbb{R}_{\geq 0}^2$, let*

$$\rho_{a,b} = (S_{a,b})_{\#}\mu, \quad S_{a,b}(x) = ax + bT(x).$$

Then

$$W_2^2(\rho_{a_1, b_1}, \rho_{a_2, b_2}) = \int \|(a_1 - a_2)x + (b_1 - b_2)T(x)\|^2 d\mu.$$

Proof. Consider the transport plan

$$\pi := (S_{a_1, b_1}, S_{a_2, b_2})_{\#}\mu \in \Pi(\rho_{a_1, b_1}, \rho_{a_2, b_2}).$$

Its transport cost is

$$\begin{aligned} & \int_{\mathbb{R}^d \times \mathbb{R}^d} \|u - v\|^2 d\pi(u, v) \\ &= \int_{\mathbb{R}^d} \|S_{a_1, b_1}(x) - S_{a_2, b_2}(x)\|^2 d\mu \\ &= \int_{\mathbb{R}^d} \|(a_1 - a_2)x + (b_1 - b_2)T(x)\|^2 d\mu. \end{aligned}$$

If we can show π is optimal, then the desired identity follows immediately. Therefore, it remains to prove that π is optimal.

By Lemma B.3, it suffices to show that the support of π

$$\Gamma := \{(S_{a_1, b_1}(x), S_{a_2, b_2}(x)) : x \in \text{supp } \mu\}$$

is c -cyclically monotone.

Let $x_1, \dots, x_m \in \text{supp } \mu$ and set $x_{m+1} = x_1$. Denote $y_i = T(x_i)$ and set $y_{m+1} = y_1$.

Define

$$u_i = a_1 x_i + b_1 y_i, \quad v_i = a_2 x_i + b_2 y_i.$$

We must show

$$\sum_{i=1}^m \|u_i - v_i\|^2 \leq \sum_{i=1}^m \|u_i - v_{i+1}\|^2.$$

Because the cost is quadratic, it is equivalent to show

$$\sum_{i=1}^m \langle u_{i+1}, v_{i+1} - v_i \rangle \geq 0.$$

Since

$$u_i = a_1 x_i + b_1 y_i, \quad v_i = a_2 x_i + b_2 y_i,$$

and hence

$$v_{i+1} - v_i = a_2(x_{i+1} - x_i) + b_2(y_{i+1} - y_i).$$

Therefore

$$\begin{aligned} & \sum_{i=1}^m \langle u_{i+1}, v_{i+1} - v_i \rangle \\ &= \sum_{i=1}^m \langle a_1 x_{i+1} + b_1 y_{i+1}, a_2(x_{i+1} - x_i) + b_2(y_{i+1} - y_i) \rangle \\ &= a_1 a_2 \Sigma_{xx} + a_1 b_2 \Sigma_{xy} + b_1 a_2 \Sigma_{yx} + b_1 b_2 \Sigma_{yy}, \end{aligned}$$

where

$$\begin{aligned} \Sigma_{xx} &:= \sum_{i=1}^m \langle x_{i+1}, x_{i+1} - x_i \rangle, & \Sigma_{xy} &:= \sum_{i=1}^m \langle x_{i+1}, y_{i+1} - y_i \rangle, \\ \Sigma_{yx} &:= \sum_{i=1}^m \langle y_{i+1}, x_{i+1} - x_i \rangle, & \Sigma_{yy} &:= \sum_{i=1}^m \langle y_{i+1}, y_{i+1} - y_i \rangle. \end{aligned}$$

We now show that each of these four quantities is nonnegative.

First, for any sequence $z_1, \dots, z_m$ with $z_{m+1} = z_1$,

$$\sum_{i=1}^m \langle z_{i+1}, z_{i+1} - z_i \rangle = \frac{1}{2} \sum_{i=1}^m \|z_{i+1} - z_i\|^2 \geq 0.$$

Applying this with $z_i = x_i$ and $z_i = y_i$ yields

$$\Sigma_{xx} \geq 0, \quad \Sigma_{yy} \geq 0.$$

Next, since T is the optimal transport map from μ to ν for quadratic cost, the graph

$$\text{Graph}(T) = \{(x, T(x))\}$$

is c -cyclically monotone. In the quadratic-cost formulation this means

$$\sum_{i=1}^m \langle x_{i+1}, T(x_{i+1}) - T(x_i) \rangle \geq 0,$$

that is,

$$\Sigma_{xy} \geq 0.$$

Moreover, because the cost $c(x, y) = \|x - y\|^2$ is symmetric, based on Lemma B.4, the flipped set

$$\{(T(x), x)\}$$

is also c -cyclically monotone. Equivalently,

$$\sum_{i=1}^m \langle T(x_{i+1}), x_{i+1} - x_i \rangle \geq 0,$$

so

$$\Sigma_{yx} \geq 0.$$

Since $a_1, b_1, a_2, b_2 \geq 0$, all coefficients in

$$a_1 a_2 \Sigma_{xx} + a_1 b_2 \Sigma_{xy} + b_1 a_2 \Sigma_{yx} + b_1 b_2 \Sigma_{yy}$$

are nonnegative. Therefore

$$\sum_{i=1}^m \langle u_{i+1}, v_{i+1} - v_i \rangle \geq 0.$$

This proves that Γ is c -cyclically monotone, and hence π is optimal.

Finally, the optimal cost is exactly the cost of π :

$$\begin{aligned} & W_2^2(\rho_{a_1, b_1}, \rho_{a_2, b_2}) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} \|u - v\|^2 d\pi(u, v) \\ &= \int_{\mathbb{R}^d} \|S_{a_1, b_1}(x) - S_{a_2, b_2}(x)\|^2 d\mu(x) \\ &= \int_{\mathbb{R}^d} \|(a_1 - a_2)x + (b_1 - b_2)T(x)\|^2 d\mu(x). \end{aligned}$$

This is the claimed formula. $\square$

B.6 FLATNESS OF THE FILLING CONE

Quadratic Form. Expanding the square in Proposition B.5,

$$\begin{aligned} W_2^2(\rho_{a_1, b_1}, \rho_{a_2, b_2}) &= (a_1 - a_2)^2 A + (b_1 - b_2)^2 C \\ &\quad + 2(a_1 - a_2)(b_1 - b_2)B. \end{aligned}$$

where $A = \int \|x\|^2 d\mu$, $B = \int \langle x, T(x) \rangle d\mu$, $C = \int \|T(x)\|^2 d\mu$.

These constants depend only on μ and T .

Proposition B.6 (Flatness of the Filling Cone). *The pullback of the W_2 -metric under $\Phi(a, b) = (S_{a, b})_{\#} \mu$ is the constant bilinear form*

$$g = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$$

in the global affine coordinates (a, b) . Consequently, the induced two-dimensional Riemannian manifold is flat, with zero Gaussian curvature.

Proof. Proposition B.5 shows that squared distances depend only on the differences $(a_1 - a_2, b_1 - b_2)$ through a fixed quadratic form. Hence the metric coefficients are constant in the (a, b) chart. A two-dimensional Riemannian metric with constant coefficients is isometric (up to a linear change of variables) to a Euclidean cone, and therefore has zero curvature. $\square$

C PROOF OF THE CLOSED-FORM EXPRESSIONS FOR THE ONE-DIMENSIONAL GAUSSIAN FAMILY

In this section, we prove Proposition 3.4 by deriving closed-form expressions for the orthogonal projection distance p , and the relative Wasserstein angle θ in one dimension.

Let

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad \nu = \mathcal{N}(0, \sigma^2),$$

where $x_1 \leq \dots \leq x_n$ are sorted samples, and consider the quadratic cost $c(x, y) = \|x - y\|^2$.

Step 1: Quantile Formula for $W_2^2(\mu, \nu)$. In one dimension, the quadratic Wasserstein distance admits the quantile representation

$$W_2^2(\mu, \nu) = \int_0^1 (F_\mu^{-1}(t) - F_\nu^{-1}(t))^2 dt. \quad (4)$$

For the empirical measure μ ,

$$F_\mu^{-1}(t) = x_i \quad \text{for } t \in \left(\frac{i-1}{n}, \frac{i}{n}\right],$$

and for $\nu = \mathcal{N}(0, \sigma^2)$,

$$F_\nu^{-1}(t) = \sigma \Phi^{-1}(t),$$

where Φ denotes the CDF of $\mathcal{N}(0, 1)$.

Substituting into Equation 4 yields

$$W_2^2(\mu, \nu) = \sum_{i=1}^n \int_{t_{i-1}}^{t_i} (x_i - \sigma \Phi^{-1}(t))^2 dt, \quad t_i = \frac{i}{n}.$$

Step 2: Change of Variables and Integrations. Let $t = \Phi(z)$, so that $dt = \phi(z) dz$, where ϕ is the standard Gaussian density. Define $z_i := \Phi^{-1}(i/n)$, with $z_0 = -\infty$, $z_n = +\infty$. Then

$$\int_{t_{i-1}}^{t_i} (x_i - \sigma \Phi^{-1}(t))^2 dt = \int_{z_{i-1}}^{z_i} (x_i - \sigma z)^2 \phi(z) dz.$$

Expanding and integrating termwise gives

$$\begin{aligned} \int_{z_{i-1}}^{z_i} (x_i - \sigma z)^2 \phi(z) dz &= \frac{x_i^2}{n} - 2\sigma x_i (\phi(z_{i-1}) - \phi(z_i)) \\ &\quad + \sigma^2 [-z\phi(z) + \Phi(z)]_{z_{i-1}}^{z_i}. \end{aligned}$$

Summing over $i = 1, \dots, n$ yields

$$W_2^2(\mu, \nu) = \frac{1}{n} \sum_{i=1}^n x_i^2 - 2\sigma \sum_{i=1}^n x_i (\phi(z_{i-1}) - \phi(z_i)) + \sigma^2. \quad (5)$$

Step 3: Computation of the Projection Length. Define the empirical scale

$$\sigma_\mu := \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2},$$

and the *projection length*

$$l := \sum_{i=1}^n x_i (\phi(z_{i-1}) - \phi(z_i)).$$

Then Equation 5 can be written as

$$W_2^2(\mu, \mathcal{N}(0, \sigma^2)) = \sigma_\mu^2 - 2\sigma l + \sigma^2.$$

Step 4: Projection Distance and RW_2 Angle. In the cone geometry of $(\mathcal{P}_2(\mathbb{R}^d)/\sim_T, RW_2)$, the projection length l , the projection distance p , and the relative angle θ satisfy the Euclidean relations

$$l = \|\mu\|_{RW_2} \cos \theta, \quad \|\mu\|_{RW_2} = \sigma_\mu.$$

Consequently,

$$\theta = \arccos\left(\frac{l}{\sigma_\mu}\right), \quad p = \sigma_\mu \sin \theta,$$

which proves Proposition 3.4.

D CLOSED-FORM EXPRESSIONS FOR OTHER LOCATION–SCALE DISTRIBUTIONS

D.1 A GENERAL FORMULA FOR ONE-DIMENSIONAL LOCATION–SCALE FAMILIES

We first present a general formula for projecting a one-dimensional distribution onto a centered location–scale family. Let μ be a zero-mean probability distribution on $\mathbb{R}$ with finite second moment, and let

$$q_\mu(u) := F_\mu^{-1}(u), \quad u \in (0, 1),$$

be its quantile function. Then

$$\|[\mu]\|_{RW_2}^2 = \int_0^1 q_\mu(u)^2 du =: \sigma_\mu^2.$$

Let ν be a centered base distribution with quantile function

$$q_\nu(u) := F_\nu^{-1}(u), \quad u \in (0, 1),$$

and assume that ν has finite nonzero second moment

$$s_\nu^2 := \int_0^1 q_\nu(u)^2 du < \infty.$$

The associated location–scale ray is given by the family $\{\nu_\sigma\}_{\sigma>0}$, whose quantile functions satisfy

$$F_{\nu_\sigma}^{-1}(u) = \sigma q_\nu(u).$$

Define the normalized base direction

$$\psi_\nu(u) := \frac{q_\nu(u)}{s_\nu}.$$

Then $\int_0^1 \psi_\nu(u)^2 du = 1$. The projection length of $[\mu]$ onto the ray generated by ν is

$$l_\nu := \int_0^1 q_\mu(u) \psi_\nu(u) du = \frac{1}{s_\nu} \int_0^1 q_\mu(u) q_\nu(u) du.$$

Consequently, the relative Wasserstein angle and the orthogonal projection distance are

$$\theta_\nu = \arccos\left(\frac{l_\nu}{\sigma_\mu}\right), \quad p_\nu = \sigma_\mu \sin \theta_\nu = \sqrt{\sigma_\mu^2 - l_\nu^2}.$$

When μ is empirical,

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad x_1 \leq \dots \leq x_n, \quad \sum_{i=1}^n x_i = 0,$$

we have

$$q_\mu(u) = x_i, \quad u \in (u_{i-1}, u_i], \quad u_i = \frac{i}{n}.$$

Thus,

$$l_\nu = \sum_{i=1}^n x_i \int_{u_{i-1}}^{u_i} \psi_\nu(u) du.$$

D.2 CLOSED-FORM EXPRESSION FOR THE UNIFORM FAMILY

We first consider the centered uniform family

$$\text{Unif}\left[-\frac{\beta}{2}, \frac{\beta}{2}\right], \quad \beta \geq 0.$$

For the base distribution $\text{Unif}[-1/2, 1/2]$, the quantile function is

$$q_{\text{unif}}(u) = u - \frac{1}{2}, \quad u \in (0, 1),$$

and

$$s_{\text{unif}}^2 = \int_0^1 \left(u - \frac{1}{2}\right)^2 du = \frac{1}{12}.$$

Hence the normalized direction is

$$\psi_{\text{unif}}(u) = \sqrt{12} \left(u - \frac{1}{2}\right).$$

The projection length of $[\mu]$ onto the uniform ray is therefore

$$\begin{aligned} l_{\text{unif}} &= \int_0^1 F_{\mu}^{-1}(u) \psi_{\text{unif}}(u) du \\ &= \sqrt{12} \sum_{i=1}^n x_i \int_{u_{i-1}}^{u_i} \left(u - \frac{1}{2}\right) du \\ &= \sqrt{12} \sum_{i=1}^n x_i \frac{2i - n - 1}{2n^2}. \end{aligned}$$

Since

$$\sigma_{\mu} = \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right)^{1/2},$$

the relative Wasserstein angle and projection distance are

$$\theta_{\text{unif}} = \arccos\left(\frac{l_{\text{unif}}}{\sigma_{\mu}}\right), \quad p_{\text{unif}} = \sigma_{\mu} \sin \theta_{\text{unif}} = \sqrt{\sigma_{\mu}^2 - l_{\text{unif}}^2}.$$

Proposition D.1 (Closed-Form Projection onto the Uniform Ray). *Let μ be a one-dimensional empirical distribution with zero mean and sorted samples $x_1 \leq \dots \leq x_n$. Then the projection length, relative Wasserstein angle, and projection distance of $[\mu]$ onto the uniform ray are given by*

$$l_{\text{unif}} = \sqrt{12} \sum_{i=1}^n x_i \frac{2i - n - 1}{2n^2},$$

and

$$\theta_{\text{unif}} = \arccos\left(\frac{l_{\text{unif}}}{\sigma_{\mu}}\right), \quad p_{\text{unif}} = \sigma_{\mu} \sin \theta_{\text{unif}}.$$

D.3 CLOSED-FORM EXPRESSION FOR THE LOGISTIC FAMILY

We next consider the centered logistic family $\text{Lg}(0, \sigma)$, $\sigma \geq 0$. The quantile function of the base logistic distribution is

$$q_{\text{Lg}}(u) = \log\left(\frac{u}{1-u}\right), \quad u \in (0, 1).$$

Since the variance of the standard logistic distribution is $\pi^2/3$, we have

$$s_{\text{Lg}} = \frac{\pi}{\sqrt{3}}, \quad \psi_{\text{Lg}}(u) = \frac{\sqrt{3}}{\pi} \log\left(\frac{u}{1-u}\right).$$

Therefore, the projection length is

$$\begin{aligned} l_{\text{Lg}} &= \int_0^1 F_{\mu}^{-1}(u) \psi_{\text{Lg}}(u) du \\ &= \frac{\sqrt{3}}{\pi} \sum_{i=1}^n x_i \int_{u_{i-1}}^{u_i} \log\left(\frac{u}{1-u}\right) du. \end{aligned}$$

Using the antiderivative

$$H_{\text{Lg}}(u) = u \log u + (1 - u) \log(1 - u),$$

with the convention $0 \log 0 = 0$, we obtain

$$l_{\text{Lg}} = \frac{\sqrt{3}}{\pi} \sum_{i=1}^n x_i [H_{\text{Lg}}(u_i) - H_{\text{Lg}}(u_{i-1})].$$

Thus,

$$\theta_{\text{Lg}} = \arccos\left(\frac{l_{\text{Lg}}}{\sigma_\mu}\right), \quad p_{\text{Lg}} = \sigma_\mu \sin \theta_{\text{Lg}} = \sqrt{\sigma_\mu^2 - l_{\text{Lg}}^2}.$$

D.4 CLOSED-FORM EXPRESSION FOR THE LAPLACE FAMILY

We finally consider the centered Laplace family $\text{Lpc}(0, b)$, $b \geq 0$. The quantile function of the base Laplace distribution with scale $b = 1$ is

$$q_{\text{Lpc}}(u) = \begin{cases} \log(2u), & 0 < u \leq \frac{1}{2}, \\ -\log(2(1 - u)), & \frac{1}{2} < u < 1. \end{cases}$$

Since the variance of $\text{Lpc}(0, 1)$ is 2, we have

$$s_{\text{Lpc}} = \sqrt{2}, \quad \psi_{\text{Lpc}}(u) = \frac{1}{\sqrt{2}} q_{\text{Lpc}}(u).$$

Therefore, the projection length is

$$\begin{aligned} l_{\text{Lpc}} &= \int_0^1 F_\mu^{-1}(u) \psi_{\text{Lpc}}(u) du \\ &= \frac{1}{\sqrt{2}} \sum_{i=1}^n x_i \int_{u_{i-1}}^{u_i} q_{\text{Lpc}}(u) du. \end{aligned} \tag{6}$$

The required antiderivative is piecewise. Define

$$H_{\text{Lpc}}(u) = \begin{cases} u \log(2u) - u, & 0 < u \leq \frac{1}{2}, \\ (1 - u) \log(2(1 - u)) - (1 - u), & \frac{1}{2} < u < 1. \end{cases}$$

Then each summand in Equation 6 is obtained from the interval endpoints:

$$\int_{u_{i-1}}^{u_i} q_{\text{Lpc}}(u) du = H_{\text{Lpc}}(u_i) - H_{\text{Lpc}}(u_{i-1}),$$

with the interval split at $u = 1/2$ whenever necessary.

Consequently,

$$\theta_{\text{Lpc}} = \arccos\left(\frac{l_{\text{Lpc}}}{\sigma_\mu}\right), \quad p_{\text{Lpc}} = \sigma_\mu \sin \theta_{\text{Lpc}} = \sqrt{\sigma_\mu^2 - l_{\text{Lpc}}^2}.$$

E THE MOMENT-MATCHING GAUSSIAN IS NOT THE W_2 -NEAREST GAUSSIAN

In this section, we explain why the moment-matching Gaussian is *not* the W_2 -nearest Gaussian approximation to a non-Gaussian distribution.

Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ have mean $\bar{\mu}$ and covariance matrix Σ_μ , and let

$$\mu_{\text{MM}} = \mathcal{N}(\bar{\mu}, \Sigma_\mu)$$

denote its moment-matching Gaussian approximation.

On the one hand, the squared length of $[\mu]$ is given by

$$\|[\mu]\|_{RW_2}^2 = RW_2^2([\delta_0], [\mu]) = \inf_{z \in \mathbb{R}^d} \int_{\mathbb{R}^d} \|x - z\|_2^2 d\mu(x).$$

The minimizer is $z = \bar{\mu}$, and hence

$$\|[\mu]\|_{RW_2}^2 = \int_{\mathbb{R}^d} \|x - \bar{\mu}\|_2^2 d\mu(x) = \text{Tr}(\Sigma_\mu).$$

On the other hand, the squared length of $[\mu_{\text{MM}}]$ is given by

$$\|[\mu_{\text{MM}}]\|_{RW_2}^2 = RW_2^2([\delta_0], [\mu_{\text{MM}}]) = W_2^2(\delta_0, \mathcal{N}(0, \Sigma_\mu)).$$

Since the squared W_2 distance from a Dirac measure to a probability measure equals the second moment of that measure about the Dirac location, we have

$$W_2^2(\delta_0, \mathcal{N}(0, \Sigma_\mu)) = \int_{\mathbb{R}^d} \|x\|_2^2 d\mathcal{N}(0, \Sigma_\mu)(x) = \text{Tr}(\Sigma_\mu).$$

Thus,

$$\|[\mu]\|_{RW_2} = \|[\mu_{\text{MM}}]\|_{RW_2} = \sqrt{\text{Tr}(\Sigma_\mu)}.$$

Now consider the Gaussian ray containing $[\mu_{\text{MM}}]$, and let θ be the RW_2 angle between this ray and the ray generated by $[\mu]$. Set

$$r := \|[\mu]\|_{RW_2} = \|[\mu_{\text{MM}}]\|_{RW_2}.$$

In the Euclidean filling cone spanned by the two rays, the orthogonal projection of $[\mu]$ onto the Gaussian ray has length

$$l = r \cos \theta.$$

Since μ is non-Gaussian, its ray differs from the Gaussian ray containing $[\mu_{\text{MM}}]$. Hence $\theta > 0$ and $\cos \theta < 1$, so

$$l = r \cos \theta < r = \|[\mu_{\text{MM}}]\|_{RW_2}.$$

This implies that the projection point of $[\mu]$ onto the Gaussian ray differs from the moment-matching Gaussian, since its length is strictly smaller than $\|[\mu_{\text{MM}}]\|_{RW_2}$. Since the projection point is a Gaussian distribution that is closer to μ than the moment-matching Gaussian is, the moment-matching Gaussian is different from the W_2 -nearest Gaussian approximation of μ .

F COORDINATE-SEPARABLE DISTRIBUTIONS IN THE PCA BASIS

Let $\mu = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ be an empirical distribution with mean $\bar{\mu}$ and covariance matrix

$$\Sigma_\mu = U \text{diag}(\sigma_1^2, \dots, \sigma_d^2) U^\top,$$

where U is orthogonal and $\sigma_k^2 > 0$ for $k = 1, \dots, d$.

Define the PCA coordinates and their marginal distributions by

$$y_i := U^\top(x_i - \bar{\mu}), \quad \mu_k := \frac{1}{n} \sum_{i=1}^n \delta_{y_{i,k}}.$$

We say that μ is *coordinate-separable in the PCA basis* if

$$(U^\top(\cdot - \bar{\mu}))_{\#} \mu = \bigotimes_{k=1}^d \mu_k.$$

For each coordinate k , sort the samples such that

$$y_{1,k} \leq \dots \leq y_{n,k},$$

and define

$$l_k := \sum_{i=1}^n y_{i,k} (\phi(z_{i-1}) - \phi(z_i)), \quad z_i := \Phi^{-1}(i/n),$$

where $z_0 = -\infty$, $z_n = +\infty$, and ϕ and Φ denote the standard normal density and cumulative distribution functions, respectively.

Consider any Gaussian distribution ν with mean $\bar{\mu}$, and let s_k^2 denote its k -th marginal variance in the PCA basis. By the one-dimensional formula,

$$W_2^2(\mu_k, \mathcal{N}(0, s_k^2)) = \sigma_k^2 - 2s_k l_k + s_k^2 = \sigma_k^2 - l_k^2 + (s_k - l_k)^2.$$

Hence, the coordinate-wise distance is minimized at $s_k = l_k$.

Moreover, any coupling of the full distributions induces, under coordinate projection, a coupling between each pair of corresponding marginals. Since the squared Euclidean transport cost decomposes as the sum of the coordinate-wise costs, we obtain

$$W_2^2(\mu, \nu) \geq \sum_{k=1}^d W_2^2(\mu_k, \mathcal{N}(0, s_k^2)) \geq \sum_{k=1}^d (\sigma_k^2 - l_k^2).$$

This lower bound is attained by choosing the Gaussian whose coordinates are independent in the PCA basis with $s_k = l_k$. Indeed, under the coordinate-separability assumption, the product of the one-dimensional optimal couplings is then an optimal coupling for the full distributions.

Since a W_2 -nearest Gaussian must have the same mean as μ , a W_2 -nearest Gaussian is therefore

$$\nu^* = \mathcal{N}(\bar{\mu}, U \text{diag}(l_1^2, \dots, l_d^2) U^\top).$$

Consequently, the minimum RW_2 angle to the Gaussian family and the corresponding projection distance are

$$\theta^* = \arccos \sqrt{\frac{\sum_{k=1}^d l_k^2}{\sum_{k=1}^d \sigma_k^2}}, \quad p^* = \sqrt{\sum_{k=1}^d (\sigma_k^2 - l_k^2)}.$$

G AN OBSTRUCTION TO EXTENDING PAIRWISE FLATNESS TO THREE RAYS

We present an example in which each pair of three distributional rays determines a unique flat filling cone, but the union of these cones, equipped with the ambient RW_2 distance, admits no Euclidean isometric embedding.

Setup. Let

$$\gamma := \mathcal{N}(0, 1) \otimes \delta_0$$

be the standard one-dimensional Gaussian distribution supported on the x -axis in $\mathbb{R}^2$. Let R_θ denote rotation about the origin by angle θ , and define

$$\mu_1 := \gamma, \quad \mu_2 := (R_{\pi/3})_\# \gamma, \quad \mu_3 := (R_{2\pi/3})_\# \gamma.$$

These are centered, degenerate Gaussian distributions supported on three lines. Each has unit RW_2 norm. Because all distributions considered below are centered, their RW_2 distances equal their W_2 distances.

Pairwise Optimal Maps. For unit vectors $u, v \in \mathbb{R}^2$ and $s, t \geq 0$, the quadratic Wasserstein distance between the corresponding line-supported Gaussians is

$$W_2^2(\mathcal{N}(0, s^2 uu^\top), \mathcal{N}(0, t^2 vv^\top)) = s^2 + t^2 - 2st|u^\top v|. \quad (7)$$

Indeed, let π be a coupling of two standard normal distributions, where $z, w \in \mathbb{R}$ denote the scalar coordinates parameterizing points szu and $t w v$ on the two supporting lines. Their transport cost is

$$\int \|szu - twv\|^2 d\pi(z, w) = s^2 + t^2 - 2st(u^\top v) \int zw d\pi(z, w).$$

By the Cauchy–Schwarz inequality and the unit second moments of both marginals, we have

$$\left| \int zw d\pi(z, w) \right| \leq \left(\int z^2 d\pi(z, w) \right)^{1/2} \left(\int w^2 d\pi(z, w) \right)^{1/2} = 1.$$

The minimum is attained by $w = z$ when $u^\top v > 0$ and by $w = -z$ when $u^\top v < 0$. For $s, t > 0$ and $u^\top v \neq 0$, the equality condition also establishes uniqueness of the optimal coupling.

Consequently, the unique optimal maps between our three line-supported distributions, $\{\mu_1, \mu_2, \mu_3\}$, restricted to their supporting lines, are

$$T_{1 \rightarrow 2} = R_{\pi/3}, \quad T_{2 \rightarrow 3} = R_{\pi/3}, \quad T_{1 \rightarrow 3} = R_{-\pi/3}.$$

The last map is valid because rotations by $-\pi/3$ and $2\pi/3$ produce the same centered Gaussian distribution on the target line, but the former gives the smaller transport cost.

Equation 7 yields

$$W_2^2(\mu_i, \mu_j) = 1, \quad i \neq j.$$

Since each measure has unit norm, all three pairwise RW_2 angles equal $\pi/3$.

Pairwise Flatness. The measures above are singular in $\mathbb{R}^2$, so we verify flatness directly.

For each pair, orient the two supporting lines according to its optimal map. The oriented unit vectors u, v then satisfy $u^\top v = 1/2$. The displacement interpolations between dilations of the two measures, including their boundary limits, have the form

$$\mathcal{N}(0, (au + bv)(au + bv)^\top), \quad a, b \geq 0.$$

The vectors $au + bv$ form a Euclidean sector of angle $\pi/3$. Any two vectors x, y in this sector have $x^\top y \geq 0$, so Equation 7 gives

$$W_2^2(\mathcal{N}(0, xx^\top), \mathcal{N}(0, yy^\top)) = \|x - y\|^2.$$

Thus each closed pairwise filling cone is isometric to a Euclidean sector and is therefore flat.

Failure of a Common Euclidean Embedding. Let ν_{12} and ν_{23} be the geodesic midpoints between μ_1, μ_2 and μ_2, μ_3 , respectively:

$$\nu_{12} := \left(\frac{\text{Id} + R_{\pi/3}}{2} \right)_{\#} \mu_1, \quad \nu_{23} := \left(\frac{\text{Id} + R_{\pi/3}}{2} \right)_{\#} \mu_2.$$

Using the rotation identity

$$\frac{\text{Id} + R_{\pi/3}}{2} = \frac{\sqrt{3}}{2} R_{\pi/6},$$

we obtain

$$\nu_{12} = \left(\frac{\sqrt{3}}{2} R_{\pi/6} \right)_{\#} \gamma, \quad \nu_{23} = \left(\frac{\sqrt{3}}{2} R_{\pi/2} \right)_{\#} \gamma.$$

Both measures have squared norm $3/4$, and their supporting lines make an angle of $\pi/3$. Therefore,

$$W_2^2(\nu_{12}, \nu_{23}) = \frac{3}{4} + \frac{3}{4} - 2 \frac{3}{4} \cos \frac{\pi}{3} = \frac{3}{4}. \quad (8)$$

Suppose that the union of the three pairwise filling cones admits an isometric embedding into a Euclidean space, and let q_i be the image of μ_i . An isometry preserves metric midpoints, so the images of ν_{12} and ν_{23} must be $(q_1 + q_2)/2$ and $(q_2 + q_3)/2$, respectively. Their squared distance would then equal

$$\left\| \frac{q_1 + q_2}{2} - \frac{q_2 + q_3}{2} \right\|^2 = \frac{1}{4} \|q_1 - q_3\|^2 = \frac{1}{4} W_2^2(\mu_1, \mu_3) = \frac{1}{4},$$

contradicting Equation 8. Hence no such Euclidean isometric embedding exists.

This example shows that flatness of every pairwise filling cone does not guarantee a common Euclidean geometry when three rays are considered together. It does not rule out flatness for special triples of rays.

H ONE EXAMPLE WHERE THE NEAREST GAUSSIAN DOES NOT SHARE EIGENVALUES OR EIGENVECTORS

This section gives a concrete example showing that the W_2 -nearest Gaussian approximation of an empirical measure may share neither the same eigenvalues nor the same eigenvectors with the moment-matching Gaussian.

H.1 A TWO-DIMENSIONAL COUNTEREXAMPLE

Consider the empirical distribution supported on three points

$$\gamma = \frac{1}{3}(\delta_{(0,0)} + \delta_{(4,0)} + \delta_{(0,3)}),$$

with mean

$$\bar{\gamma} = \left(\frac{4}{3}, 1\right).$$

Define the centered and RW_2 -normalized distribution

$$\mu := \frac{\gamma - \bar{\gamma}}{\|\gamma - \bar{\gamma}\|_{RW_2}},$$

so that $\bar{\mu} = 0$ and $\|\mu\|_{RW_2} = 1$.

Empirical Covariance and Moment Matching. A direct computation shows that the empirical covariance of μ is

$$\Sigma_\mu = \frac{1}{50} \begin{pmatrix} 32 & -12 \\ -12 & 18 \end{pmatrix}.$$

which has trace one and a nonzero off-diagonal entry.

Eigenvectors. In two dimensions, the symmetric covariance matrix Σ_μ admits an eigendecomposition of the form

$$\Sigma_\mu = R(\omega_{\text{MM}}) \Lambda R(\omega_{\text{MM}})^\top, \quad R(\omega) = \begin{pmatrix} \cos \omega & -\sin \omega \\ \sin \omega & \cos \omega \end{pmatrix},$$

where Λ is diagonal and $\omega_{\text{MM}} \in [0, \pi)$ determines the (principal axis) orientation of the eigenvectors. The corresponding rotation angle can be obtained from

$$\omega_{\text{MM}} = \frac{1}{2} \text{atan2}(2\Sigma_{\mu,12}, \Sigma_{\mu,11} - \Sigma_{\mu,22}) \pmod{\pi}.$$

Substituting the entries of Σ_μ gives

$$\omega_{\text{MM}} \approx 0.8340 \pi.$$

To determine the eigenvector orientation of the W_2 -nearest Gaussian, we perform a brute-force search over

$$\lambda \in (0, 1), \quad \omega \in [0, \pi),$$

using the parameterization

$$\Sigma(\lambda, \omega) = R(\omega) \text{diag}(\lambda, 1 - \lambda) R(\omega)^\top,$$

which represents zero-mean Gaussian covariance directions normalized to unit trace. Using Monte Carlo sampling together with exact discrete optimal transport, we obtain a minimizing Gaussian with rotation angle

$$\omega_\star \approx 0.7983 \pi.$$

Since

$$\omega_\star \neq \omega_{\text{MM}},$$

the W_2 -nearest Gaussian and the moment-matching Gaussian have different eigenvector orientations.

Eigenvalues. The discrepancy between the two Gaussian approximations is not limited to their eigenvectors. After normalizing both covariance matrices to have unit trace, their corresponding eigenvalues are

$$\begin{aligned}\text{Spec}(\text{moment-matching Gaussian}) &= \{0.7778, 0.2222\}, \\ \text{Spec}(W_2\text{-nearest Gaussian}) &= \{0.9411, 0.0589\}.\end{aligned}$$

Thus, the W_2 -nearest Gaussian is substantially more anisotropic than the moment-matching Gaussian, demonstrating that the two approximations differ not only in their principal directions but also in their covariance spectra.

Energy Landscape Visualization. Figure 3 visualizes the RW_2 energy landscape

$$(\lambda, \omega) \mapsto W_2^2(\mu, \mathcal{N}(0, \Sigma(\lambda, \omega)))$$

over Gaussian directions with unit trace. The moment-matching Gaussian and the RW_2 -nearest Gaussian are marked explicitly. The figure shows that the moment-matching direction is not a local minimizer of the RW_2 objective and that a different combination of eigenvalues and eigenvectors achieves a strictly smaller transport cost.

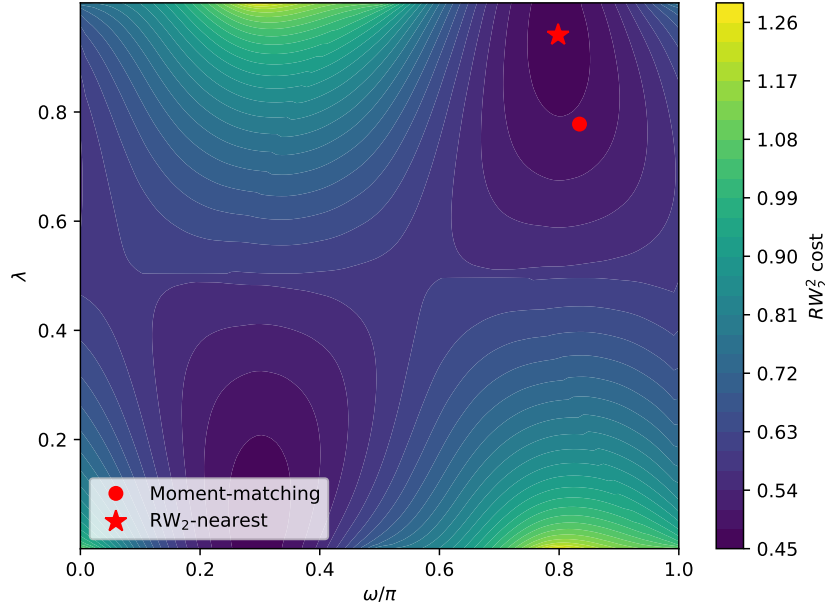


Figure 3: RW_2 energy landscape over Gaussian directions. The contour plot shows the value of $W_2^2(\mu, \mathcal{N}(0, \Sigma(\lambda, \omega)))$ as a function of the eigenvalue parameter λ and the rotation angle ω . The red dot denotes the moment-matching Gaussian, while the red star denotes the RW_2 -nearest Gaussian. The latter achieves a strictly smaller RW_2 distance, demonstrating that the moment-matching Gaussian does not lie on the nearest Gaussian ray.

I SUPPLEMENTARY EXPERIMENTAL RESULTS

I.1 VISUALIZATIONS IN THE ONE-DIMENSIONAL CASE

We visualize the difference between the W_2 -nearest Gaussian and the moment-matching Gaussian using the two-component Gaussian mixture introduced in Section 4.1. For each separation parameter $m_c \in \{1, 2, 4\}$, we draw $N = 256$ samples to construct an empirical distribution. We compute its W_2 -nearest Gaussian using the closed-form expressions and compare it with the moment-matching Gaussian. The standard deviations of the moment-matching Gaussian distributions are given by the empirical standard deviations.

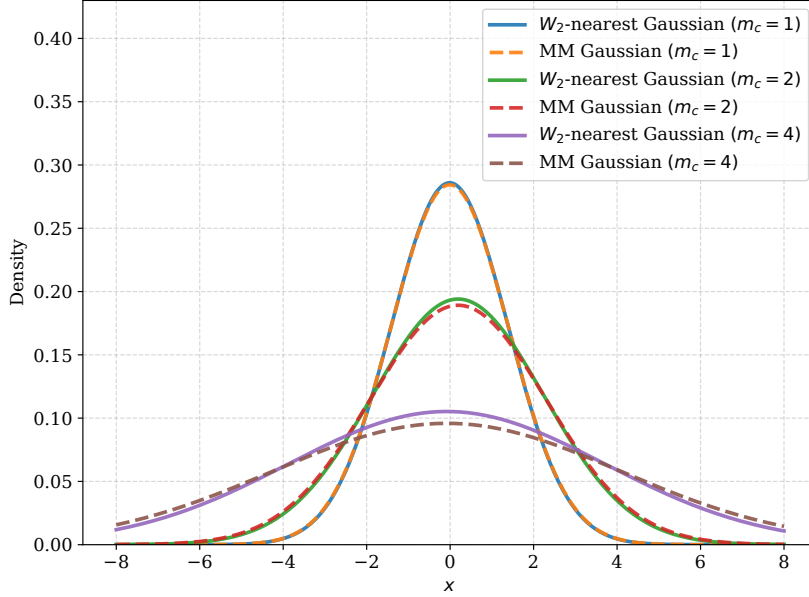


Figure 4: Comparison between the W_2 -nearest Gaussian and the moment-matching Gaussian for one-dimensional Gaussian mixture model with different separation parameters m_c . Solid curves represent the W_2 -nearest Gaussian, while dashed curves represent the moment-matching Gaussian. The two Gaussian distributions are close when the mixture components are weakly separated. However, as m_c increases, the discrepancy becomes more visible. This illustrates that the moment-matching Gaussian is not necessarily the W_2 -nearest Gaussian, even in one dimension.

Figure 4 shows that the two Gaussian distributions are close to each other when the mixture components are weakly separated. However, as m_c increases, the discrepancy becomes more pronounced. This illustrates that the moment-matching Gaussian is not necessarily the W_2 -nearest Gaussian, even in one dimension.

I.2 NON-GAUSSIANITY EVALUATION OF THE DATASETS

Setup. We evaluate both raw and latent data representations. Specifically, we use MNIST (Lecun et al., 1998) and EMNIST (Cohen et al., 2017) for evaluation in pixel space, and CIFAR-10 (Krizhevsky & Hinton, 2009) and CINIC-10 (Darlow et al., 2018) for evaluation in feature space, with features extracted by a pretrained Inception-v3 encoder (Szegedy et al., 2015).

For each dataset, we consider the full empirical distribution, individual classes, and selected mixtures of classes. We use Algorithm 2 to estimate the RW_2 angle between each distribution and its W_2 -nearest Gaussian approximation, with the moment-matching Gaussian approximation as a baseline. We report the mean and standard deviation of the estimated angles and running times. Each experiment is repeated five times with different random seeds, using a base seed of 0. We also track the RW_2 angle over optimization iterations to assess the empirical convergence and stability of the proposed algorithm.

We compute exact discrete optimal transport using `ot.emd` from the POT library (Flamary et al., 2021). To improve computational efficiency and stability, we apply PCA to reduce each representation to $d = 32$ dimensions. After PCA, we center each sampled dataset and rescale it to have unit empirical second moment. We use $n = m = 512$ and $\eta = 128$, corresponding to an effective particle step size of $\eta/m = 0.25$. We run 50 iterations and disable tolerance-based early stopping.

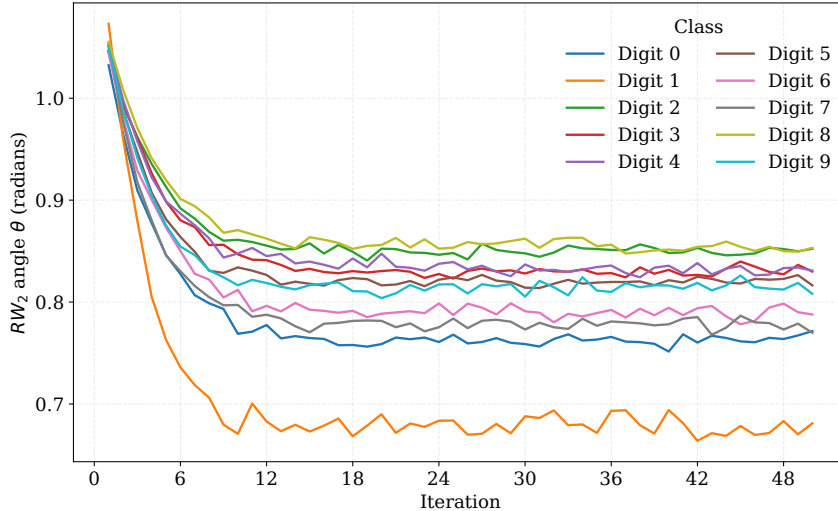


Figure 5: Evolution of the estimated RW_2 angle between each MNIST digit-class distribution and its Gaussian approximation during the first experimental repeat. Each curve corresponds to one digit class (0–9). The angles decrease rapidly during the initial iterations and subsequently stabilize around class-dependent values, providing empirical evidence of the convergence and stability of Algorithm 2.

Convergence and Stability of the Algorithm. Figure 5 shows the evolution of the estimated RW_2 angle between each MNIST digit-class distribution and its Gaussian approximation during the first experimental repeat. Each curve corresponds to one digit class (0–9). The angles decrease rapidly during the initial iterations and subsequently stabilize around class-dependent values, providing empirical evidence of the convergence and stability of Algorithm 2.

Experimental Results on MNIST and EMNIST Datasets. Tables 1 and 2 report the RW_2 angles for the full empirical distributions, individual digit classes, and selected class mixtures of MNIST and EMNIST, respectively.

For the MNIST dataset shown in Table 1, the W_2 -nearest angle for the full distribution is larger than that of most individual digit classes, indicating stronger non-Gaussianity. Across all evaluated distributions, the estimated W_2 -nearest angle yields slightly smaller angles than the moment-matching baseline, providing a better Gaussian approximation at a higher computational cost. Among all

Table 1: Non-Gaussianity evaluation of MNIST after PCA reduction to $d = 32$ dimensions. We report the mean and standard deviation of the RW_2 angles between each empirical distribution and its estimated W_2 -nearest and moment-matching Gaussian approximations, together with the running time. All angles are measured in radians.

Distribution	Classes	MM angle	W_2 -nearest angle	MM time / sec	W_2 -nearest time / sec
Full dataset	all	0.9757 ± 0.0032	0.9287 ± 0.0040	0.0403	3.2181
Digit 0	0	0.8273 ± 0.0013	0.7694 ± 0.0037	0.0312	2.9139
Digit 1	1	0.7261 ± 0.0144	0.6675 ± 0.0093	0.0274	2.6568
Digit 2	2	0.9005 ± 0.0038	0.8490 ± 0.0065	0.0288	2.9011
Digit 3	3	0.8868 ± 0.0037	0.8293 ± 0.0019	0.0285	2.8844
Digit 4	4	0.8816 ± 0.0059	0.8294 ± 0.0058	0.0299	2.8641
Digit 5	5	0.8715 ± 0.0084	0.8146 ± 0.0055	0.0295	2.8737
Digit 6	6	0.8422 ± 0.0042	0.7874 ± 0.0065	0.0284	2.8429
Digit 7	7	0.8357 ± 0.0054	0.7789 ± 0.0065	0.0285	2.8447
Digit 8	8	0.9046 ± 0.0071	0.8528 ± 0.0043	0.0298	2.9332
Digit 9	9	0.8663 ± 0.0090	0.8073 ± 0.0040	0.0289	2.8845
Mixture	0,1	0.8460 ± 0.0042	0.7971 ± 0.0063	0.0287	2.8176
Mixture	0,1,2,3	0.9281 ± 0.0061	0.8738 ± 0.0072	0.0288	2.8651
Mixture	0,1,2,3,4,5,6,7	0.9679 ± 0.0033	0.9203 ± 0.0037	0.0289	2.9212
Mixture	3,4	0.9147 ± 0.0059	0.8644 ± 0.0068	0.0298	2.9211
Mixture	3,4,5,6	0.9483 ± 0.0030	0.8983 ± 0.0043	0.0294	2.9114
Mixture	1,2,3,4,5,6,7,8	0.9755 ± 0.0031	0.9275 ± 0.0028	0.0298	2.9358
Mixture	5,6	0.9014 ± 0.0058	0.8487 ± 0.0054	0.0294	2.8877
Mixture	5,6,7,8	0.9582 ± 0.0028	0.9066 ± 0.0031	0.0293	2.9299
Mixture	2,3,4,5,6,7,8,9	0.9737 ± 0.0072	0.9320 ± 0.0030	0.0291	2.9416

Table 2: Non-Gaussianity evaluation of EMNIST after PCA reduction to $d = 32$ dimensions. We report the mean and standard deviation of the RW_2 angles between each empirical distribution and its estimated W_2 -nearest and moment-matching Gaussian approximations, together with the running time. All angles are measured in radians.

Distribution	Classes	MM angle	W_2 -nearest angle	MM time / sec	W_2 -nearest time / sec
Full dataset	all	0.9693 ± 0.0012	0.9211 ± 0.0009	0.0451	3.1928
Digit 0	0	0.8275 ± 0.0068	0.7683 ± 0.0047	0.0317	3.0877
Digit 1	1	0.7308 ± 0.0122	0.6733 ± 0.0081	0.0299	2.9363
Digit 2	2	0.9111 ± 0.0056	0.8588 ± 0.0040	0.0340	3.1503
Digit 3	3	0.8948 ± 0.0062	0.8406 ± 0.0038	0.0287	3.1221
Digit 4	4	0.8990 ± 0.0042	0.8523 ± 0.0043	0.0319	3.1737
Digit 5	5	0.8697 ± 0.0052	0.8111 ± 0.0040	0.0310	3.0895
Digit 6	6	0.8563 ± 0.0048	0.8062 ± 0.0023	0.0328	3.1696
Digit 7	7	0.8420 ± 0.0047	0.7937 ± 0.0037	0.0319	3.1014
Digit 8	8	0.9236 ± 0.0071	0.8738 ± 0.0072	0.0345	3.1455
Digit 9	9	0.8686 ± 0.0084	0.8150 ± 0.0049	0.0281	3.0975
Mixture	0,1	0.8321 ± 0.0080	0.7845 ± 0.0091	0.0310	3.0872
Mixture	0,1,2,3	0.9259 ± 0.0059	0.8733 ± 0.0078	0.0290	2.8607
Mixture	0,1,2,3,4,5,6,7	0.9617 ± 0.0050	0.9171 ± 0.0041	0.0300	2.9072
Mixture	3,4	0.9202 ± 0.0067	0.8739 ± 0.0100	0.0285	2.9214
Mixture	3,4,5,6	0.9497 ± 0.0050	0.9071 ± 0.0055	0.0288	2.9048
Mixture	1,2,3,4,5,6,7,8	0.9712 ± 0.0061	0.9278 ± 0.0051	0.0296	2.9163
Mixture	5,6	0.9230 ± 0.0048	0.8720 ± 0.0080	0.0298	2.8769
Mixture	5,6,7,8	0.9502 ± 0.0047	0.9032 ± 0.0029	0.0290	2.8985
Mixture	2,3,4,5,6,7,8,9	0.9686 ± 0.0037	0.9230 ± 0.0035	0.0287	2.9132

single-class distributions, digit 1 has the smallest angle, $\theta_{RW_2} = 0.6675$, suggesting that it is closer to a single Gaussian-like source.

Furthermore, the selected class mixtures exhibit a consistent pattern related to the distributional heterogeneity: the RW_2 angle increases from 0.7694 for digit $\{0\}$, to 0.7971 for digits $\{0, 1\}$, 0.8738 for digits $\{0, 1, 2, 3\}$, and 0.9203 for digits $\{0, 1, 2, 3, 4, 5, 6, 7\}$. A similar trend is observed for another sequence of class mixtures, progressing from $\{3\}$ to $\{3, 4\}$, then to $\{3, 4, 5, 6\}$, and finally to $\{1, 2, 3, 4, 5, 6, 7, 8\}$. These results suggest that combining more heterogeneous classes introduces

additional distributional structure that is progressively less well represented by a single Gaussian. This indicates that the RW_2 angle can potentially serve as a strong indicator of distributional heterogeneity.

For the EMNIST dataset shown in Table 2, the overall full dataset distribution has an RW_2 angle value similar to that of the MNIST dataset; however, some individual data distributions differ. For example, the RW_2 angle of digit 4 in the MNIST dataset is 0.8294 ± 0.0058 , which is smaller than that of digit 4 in the EMNIST dataset, 0.8523 ± 0.0043 . This indicates that digit 4 in the MNIST dataset is closer to a single Gaussian-like source than digit 4 in the EMNIST dataset, indicating that the former is less heterogeneous than the latter.

Overall, the RW_2 angle provides an interpretable scalar summary of dataset non-Gaussianity and reveals differences across classes and selected mixtures, and shows a stronger indication of distributional heterogeneity for raw representations.

Experimental Results on CIFAR-10 and CINIC-10 Datasets. Tables 3 and 4 report the RW_2 angles for the full empirical distributions, individual classes, and selected class mixtures of CIFAR-10 and CINIC-10, respectively.

For the CIFAR-10 dataset shown in Table 3, the W_2 -nearest angle for the full distribution is larger than that of most individual classes, indicating stronger non-Gaussianity. Across all evaluated distributions, the estimated W_2 -nearest angle yields slightly smaller angles than the moment-matching baseline, providing a better Gaussian approximation at a higher computational cost. Among all single-class distributions, the truck class has the smallest angle, $\theta = 0.7987$, suggesting that it is closer to a single Gaussian-like source.

Furthermore, the selected class mixtures exhibit a consistent pattern related to the distributional heterogeneity: the angle increases from 0.8331 for classes $\{0, 1\}$, to 0.9015 for $\{0, 1, 2, 3\}$, and further to 0.9227 for $\{0, 1, 2, 3, 4, 5, 6, 7\}$. A similar trend is observed for CINIC-10, where the corresponding angles increase from 0.8401 for classes $\{0, 1\}$, to 0.9023 for $\{0, 1, 2, 3\}$, and then to 0.9137 for $\{0, 1, 2, 3, 4, 5, 6, 7\}$. These results suggest that combining more heterogeneous classes introduces additional distributional structure that is progressively less well represented by a single Gaussian. These results suggest that the RW_2 angle can serve as a useful indicator for capturing distributional heterogeneity.

For the CINIC-10 dataset shown in Table 4, the overall full dataset distribution has an RW_2 angle value similar to that of the CIFAR-10 dataset; however, some individual data distributions differ. For example, the W_2 -nearest angle of the horse class in the CIFAR-10 dataset is 0.8159 ± 0.0089 , which is smaller than that of the horse class in the CINIC-10 dataset, 0.8527 ± 0.0078 . This indicates that the horse class in the CIFAR-10 dataset is closer to a single Gaussian-like source than the horse class in the CINIC-10 dataset, indicating that the former is less heterogeneous than the latter.

Overall, the RW_2 angle provides an interpretable scalar summary of dataset non-Gaussianity and reveals differences across classes and selected mixtures. These results suggest that the RW_2 angle can serve as a useful indicator for capturing distributional heterogeneity.

Table 3: Non-Gaussianity evaluation of CIFAR-10 dataset using pretrained Inception-v3 features reduced to $d = 32$ dimensions by PCA. We report the mean and standard deviation of the RW_2 angles to the estimated W_2 -nearest and moment-matching Gaussian approximations, together with the average running time for each method. Angles are measured in radians.

Distribution	Classes	MM angle	W_2 -nearest angle	MM time / sec	W_2 -nearest time / sec
Full dataset	all	0.9673 ± 0.0040	0.9228 ± 0.0084	0.0445	3.1869
Class airplane	0	0.9015 ± 0.0029	0.8488 ± 0.0040	0.0343	3.1542
Class automobile	1	0.8825 ± 0.0064	0.8266 ± 0.0075	0.0343	3.1504
Class bird	2	0.9385 ± 0.0057	0.8898 ± 0.0065	0.0345	3.1625
Class cat	3	0.9352 ± 0.0076	0.8874 ± 0.0056	0.0340	3.1564
Class deer	4	0.9226 ± 0.0050	0.8681 ± 0.0065	0.0336	3.1553
Class dog	5	0.9013 ± 0.0053	0.8451 ± 0.0089	0.0338	3.1396
Class frog	6	0.9148 ± 0.0074	0.8590 ± 0.0049	0.0332	3.1471
Class horse	7	0.8719 ± 0.0081	0.8159 ± 0.0089	0.0333	3.1318
Class ship	8	0.8763 ± 0.0033	0.8216 ± 0.0062	0.0341	3.1255
Class truck	9	0.8601 ± 0.0075	0.7987 ± 0.0051	0.0333	3.1118
Mixture	0,1	0.8864 ± 0.0038	0.8331 ± 0.0056	0.0335	3.1322
Mixture	0,1,2,3	0.9491 ± 0.0042	0.9015 ± 0.0079	0.0346	3.1767
Mixture	0,1,2,3,4,5,6,7	0.9716 ± 0.0053	0.9227 ± 0.0055	0.0338	3.1654
Mixture	3,4	0.9289 ± 0.0042	0.8773 ± 0.0037	0.0327	3.1188
Mixture	3,4,5,6	0.9354 ± 0.0021	0.8927 ± 0.0079	0.0339	3.1211
Mixture	1,2,3,4,5,6,7,8	0.9674 ± 0.0056	0.9183 ± 0.0057	0.0349	3.1764
Mixture	5,6	0.9156 ± 0.0078	0.8608 ± 0.0034	0.0351	3.1533
Mixture	5,6,7,8	0.9329 ± 0.0044	0.8861 ± 0.0087	0.0347	3.1736
Mixture	2,3,4,5,6,7,8,9	0.9607 ± 0.0040	0.9161 ± 0.0028	0.0340	3.1728

Table 4: Non-Gaussianity evaluation of CINIC-10 dataset using pretrained Inception-v3 features reduced to $d = 32$ dimensions by PCA. We report the mean and standard deviation of the RW_2 angles to the estimated W_2 -nearest and moment-matching Gaussian approximations, together with the average running time for each method. Angles are measured in radians.

Distribution	Classes	MM angle	W_2 -nearest angle	MM time / sec	W_2 -nearest time / sec
Full dataset	all	0.9679 ± 0.0050	0.9152 ± 0.0025	0.0401	3.0037
Class airplane	0	0.8886 ± 0.0066	0.8319 ± 0.0040	0.0300	2.9551
Class automobile	1	0.8955 ± 0.0038	0.8349 ± 0.0036	0.0304	2.9701
Class bird	2	0.9357 ± 0.0043	0.8862 ± 0.0040	0.0299	3.0161
Class cat	3	0.9383 ± 0.0049	0.8939 ± 0.0008	0.0298	2.9658
Class deer	4	0.9323 ± 0.0079	0.8809 ± 0.0091	0.0288	2.9741
Class dog	5	0.9295 ± 0.0030	0.8735 ± 0.0046	0.0294	2.9978
Class frog	6	0.9070 ± 0.0034	0.8500 ± 0.0051	0.0299	2.9766
Class horse	7	0.9044 ± 0.0064	0.8527 ± 0.0078	0.0293	2.9739
Class ship	8	0.9021 ± 0.0042	0.8524 ± 0.0042	0.0299	2.9705
Class truck	9	0.8855 ± 0.0067	0.8258 ± 0.0058	0.0299	2.9660
Mixture	0,1	0.8951 ± 0.0030	0.8401 ± 0.0075	0.0306	2.9707
Mixture	0,1,2,3	0.9552 ± 0.0087	0.9023 ± 0.0065	0.0299	2.9993
Mixture	0,1,2,3,4,5,6,7	0.9646 ± 0.0064	0.9137 ± 0.0013	0.0310	3.0034
Mixture	3,4	0.9348 ± 0.0029	0.8866 ± 0.0027	0.0303	2.9585
Mixture	3,4,5,6	0.9381 ± 0.0050	0.8932 ± 0.0052	0.0303	2.9802
Mixture	1,2,3,4,5,6,7,8	0.9616 ± 0.0056	0.9127 ± 0.0060	0.0303	2.9950
Mixture	5,6	0.9279 ± 0.0053	0.8715 ± 0.0073	0.0299	2.9909
Mixture	5,6,7,8	0.9379 ± 0.0063	0.8851 ± 0.0090	0.0302	2.9648
Mixture	2,3,4,5,6,7,8,9	0.9557 ± 0.0034	0.9085 ± 0.0024	0.0294	2.9945